Permanent Connection Guaranteed Resilient LEO Constellation for Remote Area Communications

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***Abstract*—Amid the rise of 6G space-air-ground-sea integration and accelerating deployment of low Earth orbit constellation (CON), especially for remote regions such as borders and marines lacking infrastructure, resilience CON with flexible coverage, robust connectivity, and disturbance tolerance are becoming essential for strategic communication and emergency scenarios. This study develops a layered evaluation framework covering coverage, connectivity, and resilience, based on the demand-reflective Fibonacci lattice grid-point (FLGP) model and inter-satellite link (ISL)-enabled space-based CON network graph. We introduce hybrid inclination (HyInc) series configuration and its extension HyInc-F involved continuous phase-factor gain with flexibility and compatibility. Specifically, a dual-resilience paradigm, comprising coverage and ISL resilience, is proposed to underpin communication enhancement and resilience reinforcement in remote areas. We develop a hybrid optimization framework combining non-dominated sorting genetic algorithm (NSGA)-III and a high-fidelity CON simulator with modular decoupling and mission adaptability, efficiently handling high-dimension objectives and enabling extensibility. Simulations yield valuable insights for resilience CON and highlight HyInc series' various superiority (*e.g.*, a 12% global coverage increase in certain scenarios) and framework's extensibility and transferability for diverse regional and strategic CON deployment.**

***Index Terms*—Low Earth orbit, resilience constellation design, inter-satellite link, HyInc, NSGA-III.**

I. Introduction

THE rise of space-air-ground-sea integrated network (SAGSIN) [1], which integrate satellite (SAT), airborne platform, *e.g.* uncrewed aerial vehicles (UAVs), and terrestrial infrastructure [2], aims to deliver seamless, low-latency, high-throughput services across aviation, marine, and remote areas [3]. Compared to geosynchronous orbit, low Earth orbit (LEO) SATs, with lower latency and wider bandwidth, are more suitable for global coverage and fast response in 6G scenarios, and are thus central to resilient ubiquitous networks [4], [5]. In line with this, 3GPP has incorporated LEO SATs into non-terrestrial network standards to enable SAT-terrestrial (ST) integration. This has catalyzed a global surge in large-scale LEO constellations (CONs) [6], with initiatives, *e.g. Starlink*, validating their feasibility and strategic value [7]. In particular, since 2024 and 2025, LEO constellation development has further accelerated, accompanied by expanded deployment programs and increasing adoption of AI-assisted network management. The emergence of heterogeneous and multi-orbit architectures has gradually shifted research emphasis from large-scale coverage realization toward structurally optimized and resilience-aware constellation paradigms. So as the cornerstone of SAT Internet, CON design critically influences both deployment cost and on-orbit performance [8].

Firstly, existing studies primarily target mega-CON for seamless global coverage [9], [10], [11]. For instance, [9], [10] designed ST networks with user backhaul under mega-LEO CONs, while [11] explored multi-layer schemes considering SAT mobility. Related works also optimize resource allocation [12], interference [13], latency [14], *etc.*, under mega-CON paradigm. But such mega-CON may be unaffordable or inefficient for early-stage deployment for some developing institutions [15]. Instead, customizable and affordable regional sparse CON is considered more viable [16], [17], especially for infrastructure-deficient remote areas (*e.g.* borders and marines as shown in Fig. 1), in which [17] tackled regional coverage via multi-objective optimization problem (OP).

However, a more pragmatic approach for future multi-objective OP (MOP) CONs emphasizes local enhancement atop baseline global access, especially for remote area support.

In addition, beyond coverage, inter-satellite link (ISL) modeling and connectivity have emerged as critical considerations in CON design [18]. As a key enabler of flexible and self-organized SAT networks, particularly optical ISL (OISL), is being increasingly adopted to support high-speed, decentralized space routing. Incorporating ISL and on-board regeneration (as shown in Fig. 1) allows CONs to bypass ground relays, directly forward user data among SATs, and ensure reliable communication in ground-inaccessible remote areas [19]. Relevant studies explored various ISL-centric aspects. Chen et al. analyzed phase factor impacts on ISL and propagation hops, highlighting hop minimization as a CON design strategy [20], and later proposed low-cost ISL usage metrics tied to gateway placement [21]. While [22], [23], [24] presented positive impacts of ISL in CON design phase from different perspectives, where [22] emphasized reliability via

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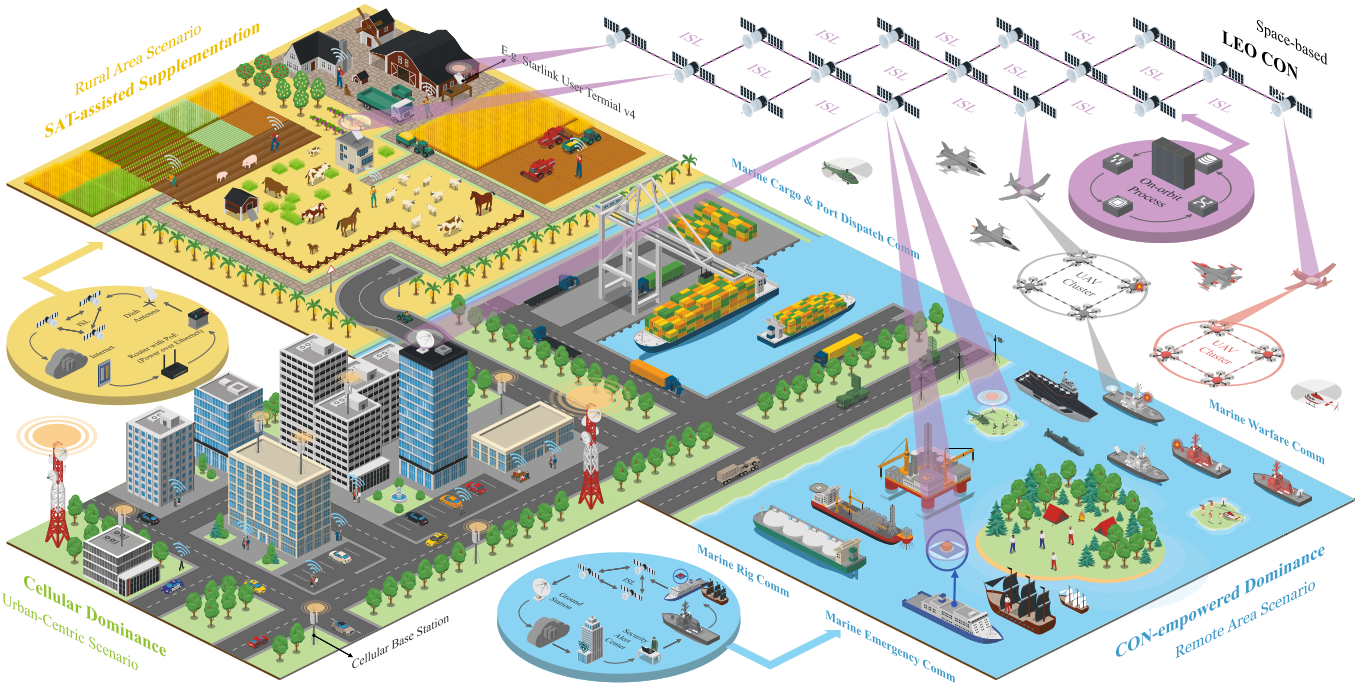


Fig. 1. SAT roles in various typical application scenarios and regenerative SAT-centric space-based LEO CON network architecture with ISL enablement.

buffer overflow mitigation, and [24] demonstrated advantages for sensor data backhauling.

However, distinct from above, we expect to further focus on challenging design of permanent strong-connected[1] CON and introduce resilience modeling directly into CON structural configuration design, which serve as a robust foundation for future network stability, and offer superior resilience against transmission disruption for constructing highly reliable communication network [19].

Besides, this study particularly emphasizes the growing resilience requirements for CON design imposed by future airspace networks, especially in remote communication contexts. As discussed above, research on resilience during the design phase remains limited, and most efforts focus on post-deployment link stability, ISL hopping, or scheduling optimization [25], [26]. While these approaches provide valuable robustness insights, they are generally evaluated on a given or pre-determined network topology. For instance, [27] enhanced ISL stability via capture probability after deployment. While in contrast to ISL redundancy optimization done in [28] and [29], there is still a lack of proactive resilience modeling framework at CON design phase, and we suggest that structural-level resilience mechanism will be vital for highly reliable CON.

In summary, to address integrated demands for coverage, connectivity, and resilience in remote scenarios, this work advances a structured paradigm for resilience-oriented CON design. Rather than evaluating robustness along a single structural dimension, this work introduces a dual-resilience paradigm that jointly models regional service continuity and constellation-level connectivity preservation within a unified configuration-sensitive framework. *First*, specifically, beyond coverage and connectivity modeling, the construction of a dual resilience framework consisting of coverage and ISL resilience, which is multi-layered, structured, and extensible, will be claimed.[2] *Second*, serving MOP collaboration, the relaxation of design constraints in traditional Walker[3] [30] or Flower [31] and the evolution of diverse CON configuration, will be advocated.[4] *Third*, for MOP collaborative optimization and deployment validation, the joint design framework coupling high-fidelity CON simulation system with evolutionary optimization, will be emphasized. MOP process will leverage Pareto dominance to balance coverage, connectivity, and resilience. Contributions summarized below:

- We highlight resilience as a configuration-sensitive structural dimension in CON design and propose a dual resilience evaluation framework, covering coverage resilience and ISL resilience under combinatorial SAT failures, that offers strong extensibility and systematically supports a stability-oriented CON design paradigm.
- We introduce a graph-based CON diagram and establish a cooperative parallel integrating coverage, connectivity, and resilience, and accordingly design a permanent connected resilience CON with global access cover and local access asset enhancement to ensure long-term sustainable service in target remote area under dynamic network.

[1]Strong and weak connection are defined by mission requirements. The former requires all SATs are always networked (either direct or indirect), while the latter is recognized by relaxing the criteria to local space-time. The permanent strong-connected topology is expected to be connected at all times.

[2]For coverage, traditional intact-state accessibility assessment falls short in reflecting reliability and service continuity, whereas the coverage guarantee capability under perturbed conditions, *e.g.* SAT failure, is expected as an important criterion for resilience. For ISL connectivity, maintaining overall connectivity amid multi-SAT failures is a key criterion.

[3]Walker Delta and Star (360° and 180° homogeneity) can be classified according to homogeneous rule of right ascension of ascending node (RAAN).

[4]Despite deployment simplicity [32], rigid constraints on altitude, inclination, and phase uniformity [33] hinder configuration freedom and adaptability.

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- We extend the hybrid inclination (HyInc) configuration to HyInc-F with continuous phase gain, offering design flexibility and backward compatibility. Theoretical analysis and simulations validate its advantages and potential as a promising option for future CON diversification.
- We develop a hybrid design optimization framework combining high-dimension evolution with high-fidelity CON simulation system, featuring sound module decoupling and mission adaptability. Moreover, the highly expandable and migratable design process flexibly supports diverse regional and strategic CON demands.

The paper is organized as follows. Section II presents system model including proposed configuration and ISL model. Section III details the evaluation framework, followed by the optimization solution in Section IV. Section V provides simulation analysis, and Section VI concludes the paper.

II. System Model

In this section, the ISL-enabled space-based network model, demand-reflective Fibonacci lattice grid point (FLGP) model, and the proposed HyInc series configuration are introduced.

A. Space-Based Network Structure and FLGP Model

The SAT's role in various typical application scenarios and regenerative SAT-centric space-based LEO CON network architecture with ISL enablement are shown in Fig. 1. Compared with non-ISL bent SAT applied in early stage, ISL-enabled regenerative SAT that incorporates on-board process (*e.g.* Chinese *Trisolaris Calculation CON*) can enable short-chain closed-loop capability and avoid experiencing multi-hop repetitive delays, thus further enhance time efficiency of the space-based network. And in order to more effectively evaluate coverage performance over target area, we introduced a grid point (GP) model [34] based on Fibonacci lattice [35] instead of traditional latitude/longitude lattice, *i.e.* FLGP, to avoid latitude weighting issue of candidate GP $\mathfrak{g} \in \mathcal{G} = \{\mathfrak{g}_1, \mathfrak{g}_2, \ldots, \mathfrak{g}_G\}$ and enhance uniform randomness [36].

Other mainstream spherical discretization schemes, such as recursive icosahedral grids and hierarchical equal area isolatitude pixelization (HEALPix), also improve uniformity. However, the FLGP model enables direct generation of an arbitrary number of sampling points with linear computational complexity and without hierarchical subdivision. Therefore, it is adopted here as a computationally efficient and sufficiently uniform solution for coverage evaluation rather than as a claim of universal optimality.

Concretely, the spatial deployment $\mathbf{v}_{\mathfrak{g}_g} = [x_{\mathfrak{g}_g}\ y_{\mathfrak{g}_g}\ z_{\mathfrak{g}_g}]$ of GP $\mathfrak{g}_g \in \mathcal{G}, g = \{1, 2, \ldots, G\}$ in World Geodetic System-1984 (WGS-84) coordinate system follows the Fibonacci lattice GP (FLGP) generation rules as shown below:

$$x_{\mathfrak{g}_g} = [\cos(2\pi g \phi_{\text{gold}}) \sqrt{1 - [(2g-1)/G - 1]^2}] \cdot R_E, \quad (1a)$$

$$y_{\mathfrak{g}_g} = [\sin(2\pi g \phi_{\text{gold}}) \sqrt{1 - [(2g-1)/G - 1]^2}] \cdot R_E, \quad (1b)$$

$$z_{\mathfrak{g}_g} = [(2g-1)/G - 1] \cdot R_E, \quad (1c)$$

where G and ϕ_{gold} are total GP number and golden ratio factor $(\sqrt{5}-1)/2$, and R_E is the radius of Earth.

B. Walker-Derived HyInc Series CON Model

1) Orbital Elements and Recurrent Orbit: Consider a CON with all SATs orbiting in P orbital planes at altitude $h_{\mathfrak{s}}$, with circular orbits $\mathfrak{p}_p \in \mathcal{P}, p = \{1, 2, \ldots, P\}$ deployed by $S^p (S^1 = S^2 = \cdots = S^P)$ SATs uniformly in the plane in Walker background. The CON scale satisfies $S = \sum_{p=1}^{P} S^p = S^P \cdot P$ and the instantaneous position of SAT $\mathfrak{s}_s \in \mathcal{S}, s = \{1, 2, \ldots, S\}$ is localized by the six orbital elements, *i.e.*, *i)* inclination i ($\in [0°, 90°]$ for energy conservation) and RAAN longitude Ω firstly establishes SAT's motion plane; *ii)* semi-axis $a_{\mathfrak{s}}$, eccentricity e and perigee amplitude w combine to define the specific orbit shape, and have $e = 0$ and $a_{\mathfrak{s}} \overset{e=0}{\rightarrow} r_{\mathfrak{s}} = ||\mathbf{v}_{\mathfrak{s}}||_2 = R_E + h_{\mathfrak{s}}$ in Walker, where $r_{\mathfrak{s}}$ is orbit radius and $||\cdot||_2$ is the Euclidean norm; *iii)* SAT specific moment positioning depends on time-varying elements, *i.e.* the anomalies including true anomaly τ_A, eccentric anomaly E_A, and mean anomaly M_A. τ_A and E_A are associated with orbital and auxiliary SAT, respectively, and the latter are in auxiliary orbit with radius $a_{\mathfrak{s}}$ and velocity $\overline{w_v}$ (orbital SAT average angular velocity). Off nadir angle $\beta/2 = \arcsin[R_E/(h_{\mathfrak{s}} + R_E) \cdot \cos\theta_{\min}]$ relates to the minimum SAT communication elevation $\theta_{\min}$ (10°-15°). As for recurrent orbit height $H_{z_2}^{z_1}$, given $T_{\mathfrak{s}} z_2 = T_E z_1$, we have $h_{\mathfrak{s}} \rightarrow H_{z_2}^{z_1} \approx \sqrt[3]{\frac{G_{\text{Kep}} M_E}{4\pi^2} T_E^2 \left(\frac{z_1}{z_2}\right)^2} - R_E, z_1, z_2 \in \mathbb{N}^+$, where $T_E, T_{\mathfrak{s}}$ related to Earth and SAT period, $G_{\text{Kep}}, M_E, \mathbb{N}^+$ denote gravitational constant, Earth mass, and positive integer set.

2) Proposed HyInc Series Configuration: Walker can be described by parameter set $(S/P/F_\omega : i)$ with fixed height, where phase factor F_ω satisfies $F_\omega \in \{0, 1, \ldots, P-1\}$. Let Ω^0 and M_A^0 be the RAAN longitude and initial mean anomaly of the reference SAT, then RAAN longitude $\Omega^{s,p}$ of SAT $\mathfrak{s}_s^p \in \mathcal{S}^p = \{\mathfrak{s}_1^p, \mathfrak{s}_2^p, \ldots, \mathfrak{s}_{S^P}^p\}$ in orbit $\mathfrak{p}_p \in \mathcal{P}, p = \{1, 2, \ldots, P\}$ can be given by eq. (2):

$$\Omega_{\Delta}^{s,p} = \left[\Omega^0 + (p-1)\frac{2\pi}{P}\right] - \left\lfloor \frac{P\Omega^0 + 2\pi(p-1)}{2\pi} \right\rfloor 2\pi, \quad (2a)$$

$$\Omega_{\star}^{s,p} = \left[\Omega^0 + (p-1)\frac{\pi}{P}\right] - \left\lfloor \frac{P\Omega^0 + \pi(p-1)}{\pi} \right\rfloor \pi, \quad (2b)$$

where $\lfloor \cdot \rfloor$ is downward rounding operation, eq. (2a) and eq. (2b) correspond to Walker Delta ($i < 90°$) and Walker Star ($i = 90°$) cases, respectively. $M_A^{s,p}$ is given by eq. (3):

$$M_A^{s,p} = \left\{ \left[M_A^0 + (p-1) \cdot \Delta f\right] + (s^p - 1) \cdot \frac{360}{S^P} \right\} - \left\lfloor \frac{\left\{ \left[M_A^0 + (p-1) \cdot \Delta f\right] + (s^p - 1) \cdot \frac{360}{S^P} \right\}}{360} \right\rfloor \cdot 360, \quad (3)$$

where Δf denotes phase difference and can be computed by

$$\Delta f = (360/S) \cdot F_\omega, F_\omega \in \{0, 1 \ldots, P-1\}. \quad (4)$$

With generality and diversity, the proposed HyInc series relaxes the uniformity constraints from an inclination perspective as shown in Fig. 2, and the set of Pareto-optimal solutions

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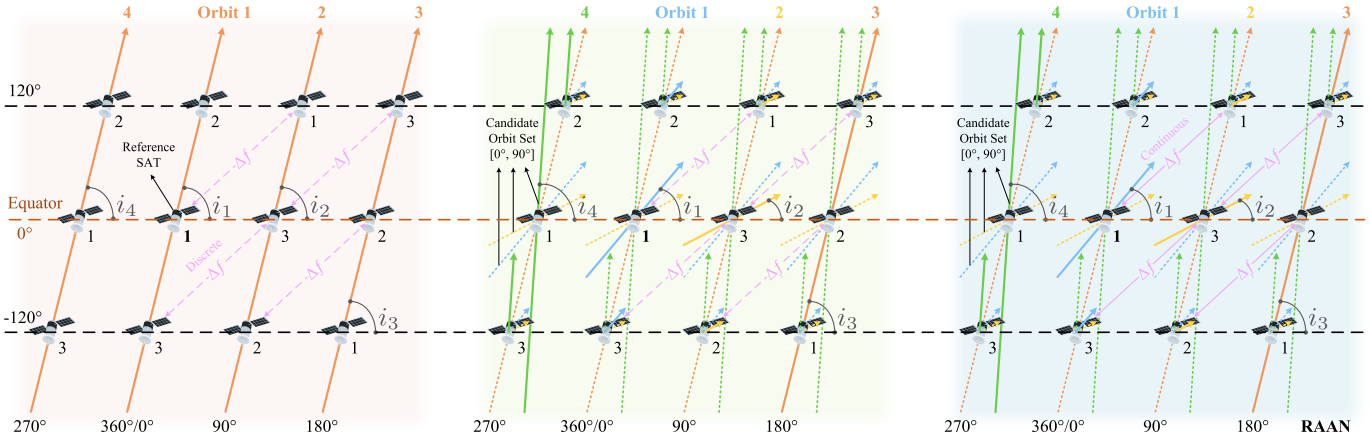


Fig. 2. Distribution diagrams of CON orbital planes in Walker, HyInc, and HyInc-F configuration, from left to right respectively.

corresponding to HyInc definitely weakly dominates Walker's from a holistic manner [37], *i.e.* compatibility. Specifically, compared to eq. (2), $\Omega^{s,p}$ under HyInc is given by eq. (5):

$$\Omega_H^{s,p} = \left[\Omega^0 + (p-1)\frac{\rho\pi}{P}\right] - \left\lfloor \frac{P\Omega^0 + \rho\pi(p-1)}{\rho\pi} \right\rfloor \rho\pi, \quad (5)$$

where $\rho = 2 - \bigvee_{p=1}^{P} \lfloor i_p/90 \rfloor, i_p \in [0°, 90°]$. Among which, $\bigvee$ is disjunction symbol and i_p denotes inclination of $\mathfrak{p}_p$. Similarly, HyInc can also be described by relevant set $(S/P/F : \mathbf{I}), \mathbf{I} = [i_1\ i_2\ \cdots\ i_P] \in \mathbb{I}^{1\times O}$ and the inclination set $\mathbb{I}$ can be further extended to $[90°, 180°]$ for retrograde orbit. It is worth noting that ρ serves as a symmetry selector unifying Delta- and Star-type RAAN equalization rules, rather than defining the hybrid property itself. The hybrid characteristic arises from the heterogeneous inclination vector $\mathbf{I}$. When all orbital planes share homogeneous inclinations ($i_p < 90°$), $\rho = 2$ and the formulation degenerates to the Walker Delta case, which is intentionally embedded as a special case within the HyInc framework. Furthermore, for configuration application considered in this study, we further propose and explore the gain space of continuous phase factor, *i.e.* consider liberalizing the constraint on discrete phase factor and further extending eq. (4) to the following continuous form:

$$\Delta f = (360/S) \cdot F_\omega, F_\omega \in [0, P-1], \quad (6)$$

and call the relevant candidate the HyInc-F configuration.

C. ISL Communication and CON Graph Model

Considering the CON Graph $\Gamma = \{\mathcal{S}, \mathcal{L}\}$ consisting of SAT set $\mathcal{S}$ and ISL set $\mathcal{L}$, the following unweighted CON adjacency matrix $\mathbf{L} = [\mathrm{L}_{ij}] \in \{0,1\}^{S\times S}$ characterizes the CON diagram Γ, where $\mathrm{L}_{ij} \in \{0,1\}$ (short for $\mathrm{L}_{\mathfrak{s}_i,\mathfrak{s}_j}$) can indicate the space-to-space link availability between SAT $\mathfrak{s}_i$ and $\mathfrak{s}_j$ in outer-space vacuum conditions, and $\mathbf{L}$ is symmetric. $\mathbf{L}$ is available by $\mathbf{L} = \mathrm{mod}(\mathbf{Pr}^+, 2) + \mathbf{E}_{S\times S}$, and $\mathbf{Pr}^+$ equals to

$$\left\lfloor \frac{\overbrace{\begin{bmatrix} \mathrm{Pr}_{11} & \mathrm{Pr}_{12} & \cdots & \mathrm{Pr}_{1S} \\ \mathrm{Pr}_{21} & \mathrm{Pr}_{22} & \cdots & \mathrm{Pr}_{2S} \\ \vdots & \vdots & \ddots & \vdots \\ \mathrm{Pr}_{S1} & \mathrm{Pr}_{S2} & \cdots & \mathrm{Pr}_{SS} \end{bmatrix}}^{\mathbf{Pr}} \odot \overbrace{\begin{bmatrix} 0 & 1 & \cdots & 1 \\ 1 & 0 & \cdots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \cdots & 0 \end{bmatrix}}^{\mathbf{M}_{S\times S}}}{\mathrm{Pr}_{\min}} \right\rfloor, \quad (7)$$

where $\mathbf{Pr} \in \mathbb{R}^{S\times S}$ is CON signal matrix, $\mathbb{R}$ is real number set and Pr_{ij} (short for $\mathrm{Pr}_{\mathfrak{s}_i,\mathfrak{s}_j}$) denotes signal power received by SAT $\mathfrak{s}_j$ from $\mathfrak{s}_i$, $\odot$ denotes Hadamard product, $\mathrm{Pr}_{\min}$ is signal power threshold, $\mathbf{M}_{S\times S} = \mathbf{J}_{S\times S} - \mathbf{E}_{S\times S}$ is mask matrix that masks own signals and only handles signals between different SATs, $\mathbf{J}_{S\times S}$ and $\mathbf{E}_{S\times S}$ are one matrix and unit matrix. Consider initial ideal case with $\mathrm{Pr}_{ij} \approx \mathrm{Pr}_{ji}$, *i.e.* $\mathbf{Pr}$ is also symmetric. Using $G_{\mathfrak{s}_i}, G_{\mathfrak{s}_j}$ to denote antenna gains of $\mathfrak{s}_i$, $\mathfrak{s}_j$, the ISL instantaneous received power can be given by

$$\mathrm{Pr}_{\mathfrak{s}_i,\mathfrak{s}_j}^{(t)} = \mathrm{Pt}_{\mathfrak{s}_i}^{(t)} + \mathrm{PL}_{\mathfrak{s}_i,\mathfrak{s}_j}^{(t)} + 10\mathrm{lg}(G_{\mathfrak{s}_i} G_{\mathfrak{s}_j}), \quad (8)$$

where PL and $\mathrm{Pt}_{\mathfrak{s}_i}$ denote path loss and transmit power, and the former calculation can follow Friis free space transmission model under space vacuum propagation conditions:

$$\mathrm{PL}_{\mathfrak{s}_i,\mathfrak{s}_j}^{(t)} = -10\mathrm{lg}\{\lambda_{\mathfrak{s}_1,\mathfrak{s}_2}^2 / [(4\pi)^2 ||\mathbf{d}_{\mathfrak{s}_i,\mathfrak{s}_j}^{(t)}||_2^2]\}, \quad (9)$$

where the wavelength $\lambda_{\mathfrak{s}_i,\mathfrak{s}_j} = c/f_{\mathfrak{s}_i,\mathfrak{s}_j}$, c and $f_{\mathfrak{s}_i,\mathfrak{s}_j}$ refer to light speed and signal frequency, and $\mathbf{d}_{\mathfrak{s}_i,\mathfrak{s}_j}^{(t)} = \mathbf{v}_{\mathfrak{s}_j}^{(t)} - \mathbf{v}_{\mathfrak{s}_i}^{(t)}$ denotes the vector of SAT $\mathfrak{s}_i$ pointing to $\mathfrak{s}_j$ at moment t.

Given the outer-space free-space propagation environment, instantaneous fading is associated only with $||\mathbf{d}_{\mathfrak{s}_i,\mathfrak{s}_j}^{(t)}||_2$ (can short to $||\mathbf{d}_{ij}^{(t)}||_2$), *i.e.* $\mathrm{Pr} = \mathcal{F}(||\mathbf{d}||_2)$. Through ISL light-of-sight (LOS) distance threshold $||\mathbf{d}_{\mathfrak{s}_i,\mathfrak{s}_j}||_2^{\max}$ (6000km considered based on OISL and latest engineering progress), $\mathbf{L}$ can then be further computed via CON distance matrix $\mathbf{D} \in \mathbb{R}^{S\times S}$, *i.e.* $\mathbf{L} = \left\lceil \frac{1}{2}\left\{ \mathrm{sgn}\left[\|\mathbf{d}_{\mathfrak{s}_i,\mathfrak{s}_j}\|_2^{\max} \cdot \mathbf{J}_{S\times S} - \mathscr{F}^{-1}(\mathbf{Pr}) \right] + 1 \right\} \right\rceil$, where $\mathbf{D} = \mathscr{F}^{-1}(\mathbf{Pr})$, $\mathscr{F}^{-1}(\cdot)$ is inverse function related to eq. (8) and eq. (9), $\lceil\cdot\rceil$ and $\mathrm{sgn}(\cdot)$ are round-up operation and sign function. It should be noted that detailed physical-layer impairments (*e.g.*, Doppler effects, pointing constraints, implementation losses) may affect link margins but do not alter the geometric feasibility structure underlying the adjacency construction in this work.

The definition associated with connected CON is as follows.

Definition 1 (Permanent Connected CON): For two SATs $\mathfrak{s}_i, \mathfrak{s}_j \in \mathcal{S}(i \neq j)$ in CON graph $\Gamma = \{\mathcal{S}, \mathcal{L}\}$, they are connected if there exists a node sequence $\mathfrak{s}_{i1}, \mathfrak{s}_{i2}, \ldots, \mathfrak{s}_{ik}$ such that $(\mathfrak{s}_i, \mathfrak{s}_{i1}), (\mathfrak{s}_{i1}, \mathfrak{s}_{i2}), \ldots, (\mathfrak{s}_{ik}, \mathfrak{s}_j) \in \mathcal{L}$. And if any two SATs are connected, CON becomes connected CON, *i.e.* satisfies $\forall \mathfrak{s}_i^{(t)}, \mathfrak{s}_j^{(t)} \in \mathcal{S}(i \neq j), \mathcal{Q}(\mathfrak{s}_i^{(t)}, \mathfrak{s}_j^{(t)}) \neq \varnothing, \forall t \in \mathcal{T}$.

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Connected CON equals strong connection CON, which is directly or indirectly reachable between any two SATs. Then the following Lemma and associated corollary can be derived.

Lemma 1: If there exists a positive integer $k < S$ (*i.e.* $k < |\mathcal{S}|$) such that $(\mathbf{L}^k)_{ij} > 0$, *i.e.* $\exists k \in \mathbb{N}, 1 \leq k < |\mathcal{S}|$, s.t. $(\mathbf{L}^k)_{ij} > 0$, then SATs $\mathfrak{s}_i$ and $\mathfrak{s}_j$ are connected.

Proof: See Appendix A. ∎

Corollary 1: If there exists a positive integer $k < S$ (*i.e.* $k < |\mathcal{S}|$) such that $(\mathbf{L}^k)_{ij} > 0$ holds for any $i \neq j$, *i.e.*,

$$\exists k \in \mathbb{N}, 1 \leq k < |\mathcal{S}|, \text{ s.t. } \forall i \neq j, (\mathbf{L}^k)_{ij} > 0, \tag{10}$$

then the corresponding CON is the connected CON.

Proof: Any $i \neq j$ satisfies Lemma 1, then any two SATs are connected and satisfy the connected CON definition. ∎

III. Performance Metrics Analysis

In this section, we present an evaluation framework that incorporates coverage (including global access and local enhancement in remote areas), ISL connectivity, and resilience.

A. Global Access Cover $\mathcal{C}_\mathcal{G}$ Analysis

Global coverage performance is considered as CON's ability to provide feasible SAT access for GPs over space and time, and is represented numerically as average access coverage of all GPs, *i.e.* $\mathcal{C}_\mathcal{G}$ (hereafter also called *global access cover*). From communication perspective, optimizing $\mathcal{C}_\mathcal{G}$ means improving service reachability and coverage continuity by providing access opportunities to more users over more time.

Use the binary variable $b^{(t)}_{\mathfrak{g},\mathfrak{s}}$ to indicate whether the communication link between GP $\mathfrak{g}$ and SAT $\mathfrak{s}$ is available at moment $t \in \mathcal{T} = \{1, 2, \ldots, T\}$, then the Lemma 2 is involved with it.

Lemma 2: The availability of ST link under circular orbit can be deduced via LOS distance threshold $||\mathbf{d}_{\mathfrak{g},\mathfrak{s}}||_2^{\max}$, *i.e.* $b^{(t)}_{\mathfrak{g},\mathfrak{s}} = \left\lceil \frac{1}{2}\left\{\text{sgn}\left[||\mathbf{d}_{\mathfrak{g},\mathfrak{s}}||_2^{\max} - ||\mathbf{d}^{(t)}_{\mathfrak{g},\mathfrak{s}}||_2\right] + 1\right\}\right\rceil$, where $\mathbf{d}_{\mathfrak{g},\mathfrak{s}} = \mathbf{v}_\mathfrak{s} - \mathbf{v}_\mathfrak{g}$ is the vector from GP $\mathfrak{g}$ to SAT $\mathfrak{s}$ and $|\mathbf{d}_{\mathfrak{g},\mathfrak{s}}||_2^{\max} = \sqrt{C_E^2 + h_\mathfrak{s}^2 + 2R_E h_\mathfrak{s}} - C_E$, $C_E = R_E \sin(\theta_{\min})$.

Proof: See Appendix B. ∎

Thus, the double-hop propagation delay $\Delta\tau_\mathfrak{g}$ satisfies

$$\Delta\tau_\mathfrak{g} \leq 2\Delta\tau_{\max} = \frac{2}{c} \cdot \sqrt{C_E^2 + h_\mathfrak{s}^2 + 2R_E h_\mathfrak{s}} - C_E, \tag{11}$$

where $\Delta\tau_{\max}$ denotes the maximum single-hop delay. The disjunction operation $b^{(t)}_\mathfrak{g} = \bigvee_{s=1}^{S} b^{(t)}_{\mathfrak{g},\mathfrak{s}_s}, \forall t \in \mathcal{T}, \mathfrak{g} \in \mathcal{G}$ informs whether GP $\mathfrak{g}$ can obtain access service at moment t, then we have access coverage ratio $\mathcal{C}_\mathfrak{g} = \frac{1}{T}\sum_{t=1}^{T} b^{(t)}_\mathfrak{g} = \frac{1}{T}\sum_{t=1}^{T}\bigvee_{s=1}^{S} b^{(t)}_{\mathfrak{g},\mathfrak{s}_s}, \forall \mathfrak{g} \in \mathcal{G}$ and finally $\mathcal{C}_\mathcal{G}$ can be defined as

$$\mathcal{C}_\mathcal{G} \triangleq \frac{1}{G}\sum_{\mathcal{G}} \mathcal{C}_\mathfrak{g} = \frac{1}{GT}\sum_{\mathcal{G}}\left[\sum_{t=1}^{T}\left(\bigvee_{s=1}^{S} b^{(t)}_{\mathfrak{g},\mathfrak{s}_s}\right)\right]. \tag{12}$$

B. Local-Enhanced Access Asset $\mathcal{A}_{\mathcal{G}_0}$ Analysis

Compared to *global access cover* $\mathcal{C}_\mathcal{G}$ analyzed in previous section, for target remote area where SAT communication is more required to supplement, we introduce *local access asset* $\mathcal{A}_{\mathcal{G}_0}$ to characterize localized communication enhancement capability, and is defined as average available SAT of GP $\mathfrak{g}_g \in \mathcal{G}_0, g = \{1, 2, \ldots, G_0\}$, where $\mathcal{G}_0$ denotes local GP set. From communication perspective, optimizing *local access asset* means improving access diversity, link selection flexibility, and potential service robustness for local users over more time.

The cooperative application of ISL and diversity techniques implies that more available SATs can further improve communication tolerance. Focusing on the set of available SATs $\mathcal{S}^{(t)}_\mathfrak{g} = \{\mathfrak{s}_1, \mathfrak{s}_2, \ldots, \mathfrak{s}_{S^{(t)}_\mathfrak{g}}\}$ of GP $\mathfrak{g}$, $S^{(t)}_\mathfrak{g}$ can be obtained by

$$S^{(t)}_\mathfrak{g} = \textstyle\sum_\mathcal{S} b^{(t)}_{\mathfrak{g},\mathfrak{s}_s}, \ \forall t \in \mathcal{T}, \mathfrak{g} \in \mathcal{G}_0. \tag{13}$$

Then we have average access asset $\mathcal{A}_\mathfrak{g} = \frac{1}{T}\sum_{t=1}^{T} S^{(t)}_\mathfrak{g} = \frac{1}{T}\sum_{t=1}^{T}\sum_{s=1}^{S} b^{(t)}_{\mathfrak{g},\mathfrak{s}_s}, \forall \mathfrak{g} \in \mathcal{G}_0$, finally $\mathcal{A}_{\mathcal{G}_0}$ can be defined as

$$\mathcal{A}_{\mathcal{G}_0} \triangleq \frac{1}{G_0}\sum_{\mathcal{G}_0}\mathcal{A}_\mathfrak{g} = \frac{1}{G_0 T}\sum_{\mathcal{G}_0}\left[\sum_{t=1}^{T}\left(\sum_{s=1}^{S} b^{(t)}_{\mathfrak{g},\mathfrak{s}_s}\right)\right]. \tag{14}$$

C. Local-Enhanced Access Resilience $\mathcal{R}^o_{\mathcal{G}_0}$ Analysis

The resilience CON analysis framework integrates ST and ISL dimensions. This section dedicates to *local access resilience* for local area from ST dimension, *i.e.* GP-centric perspective. The *CON ISL resilience* related to ISL dimension, *i.e.* CON-centric perspective, is scheduled in Section III-E.

Metric $\mathcal{R}^o_{\mathcal{G}_0}, o \in \{1, 2, \ldots, G_0\}$ is defined to characterize the ST resilience performance of local areas $\mathcal{G}_0$ when SATs fail due to malfunction, destruction, or other factors. Intuitively, $\mathcal{R}^o_{\mathcal{G}_0}, o \in \{1, 2, \ldots, G_0\}$ measures how much service capability (*i.e.*, accessible SATs) remains available to ground points after o SAT failures. And $\mathcal{R}^1_{\mathcal{G}_0}$ is the 1st*-order local access resilience* metric, which reflects the access resilience of single-SAT failure. $\mathcal{R}^o_{\mathcal{G}_0}$ implies the expectation of residual amount of $\mathcal{A}_{\mathcal{G}_0}$ when SAT failure happens, which also reflects the access asset fluctuation. Consider o SATs failed and the set of failed SAT is denoted as $\mathcal{S}^o \in \mathbb{S}^o$, where $\mathbb{S}^o = \{\mathcal{S}^o \subset \mathcal{S} \mid |\mathcal{S}^o| = o\}$ denotes failure combinations. Then available SAT set of GP $\mathfrak{g}$ gets $(\mathcal{S} \setminus \mathcal{S}^o)^{(t)}_\mathfrak{g}$, *i.e.* $|(\mathcal{S} \setminus \mathcal{S}^o)^{(t)}_\mathfrak{g}| = \sum_{\mathcal{S}\setminus\mathcal{S}^o} b^{(t)}_{\mathfrak{g},\mathfrak{s}}, \ \forall t \in \mathcal{T}, \mathfrak{g} \in \mathcal{G}_0$, the residual access asset expectation $\mathcal{R}^o_\mathfrak{g} = \frac{1}{|\mathbb{S}^o|T}\sum_{\mathbb{S}^o}\left[\sum_{t=1}^{T}\left(\sum_{\mathcal{S}\setminus\mathcal{S}^o} b^{(t)}_{\mathfrak{g},\mathfrak{s}}\right)\right], \forall \mathfrak{g} \in \mathcal{G}_0$, thus defining $\mathcal{R}^o_{\mathcal{G}_0}$ as

$$\mathcal{R}^o_{\mathcal{G}_0} \triangleq \frac{1}{\mathcal{G}_0|\mathbb{S}^o|T}\sum_{\mathcal{G}_0}\left\{\sum_{\mathbb{S}^o}\left[\sum_{t=1}^{T}\left(\sum_{\mathcal{S}\setminus\mathcal{S}^o} b^{(t)}_{\mathfrak{g},\mathfrak{s}}\right)\right]\right\}. \tag{15}$$

Simulations focus on the engineering-typical 1st*-order local access resilience*. $\mathcal{R}^o_{\mathcal{G}_0}$ uses equal-weight failure, while non-uniform weights can be incorporated when mission-specific risk priors are available, *e.g.* heterogeneous multi-layer CON.

The metric computation involved in eq. (15) requires traversing each SAT failure combination and repeating high-order nested CON simulation system operations. The complexity grows multiplicatively for larger S, $\mathcal{G}_0$, and T, and is especially costly in higher-order resilience metrics. Therefore, we propose an acceleration strategy based on offline cache of temporal ST connection matrix, following the

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principle of *space-for-time*. The corresponding lemma is as follows.

Lemma 3: Let $\mathbf{B} = [\mathbf{B}^{(1)}\ \mathbf{B}^{(2)}\ \cdots\ \mathbf{B}^{(T)}]$ be the temporal ST connection matrix, then $\mathcal{R}^{o}_{\mathcal{G}_0}$ can be directly obtained by $\mathcal{R}^{o}_{\mathcal{G}_0} = (S-o)/(STG_0)\cdot\langle \mathbf{B}, \mathbf{J}_{G_0\times TS}\rangle$, where $\mathbf{B}^{(t)} \in \{0,1\}^{G_0\times S}$ denotes ST connection matrix at moment t and $\langle\cdot,\cdot\rangle$ denotes the Frobenius inner product operation, thus we have $\mathcal{R}^{1}_{\mathcal{G}_0} = (S-1)/(STG_0)\cdot\langle \mathbf{B}, \mathbf{J}_{G_0\times TS}\rangle$ when $o=1$.

Proof: See Appendix C. ∎

Caching $\mathbf{B}$ significantly reduces simulation engine calls, decouples analysis from operations, and enables high parallelism.

D. CON ISL Connectivity $\mathcal{N}_{\mathcal{S}}$ Analysis

Since CON structure in real mission scenarios often faces multiple non-ideal factors, *e.g.* insufficient CON scale, incomplete link deployment, asymmetric arrangement, local link failure, *etc.*, the binary criterion of *connected CON or not* fails to fully characterize CON ISL structure. Therefore, the CON connected component size, *i.e.* maximal connected sub-CON scale (hereafter also called *CON ISL connectivity* $\mathcal{N}_{\mathcal{S}}$), is introduced. We have $\mathcal{N}_{\mathcal{S}}^{(t)} = \max\{\mathbf{L}^{(t)}_{\text{reach}}\mathbf{J}_{S\times 1}\}, \forall t\in\mathcal{T}, 1\le \mathcal{N}_{\mathcal{S}}\le S$, where $\mathbf{L}^{(t)}_{\text{reach}} = \text{sgn}(\mathbf{L}^{(t)}_{\text{exist}}) \in \{0,1\}^{S\times S}$ is the ISL reachability matrix at moment t and satisfies $\mathbf{L}^{(t)}_{\text{reach}}(i,j) = (\mathfrak{s}_i$ reachable to $\mathfrak{s}_j)$? 1: 0, and $\mathbf{L}^{(t)}_{\text{exist}} = \sum_{k=1}^{S-1}(\mathbf{L}^{(t)})^k$ is the ISL existence matrix. Then $\mathcal{N}_{\mathcal{S}}$ can be finally defined as

$$\mathcal{N}_{\mathcal{S}} \triangleq \frac{1}{T}\sum_{t=1}^{T}\mathcal{N}_{\mathcal{S}}^{(t)} = \frac{1}{T}\sum_{t=1}^{T}\max\left\{\mathbf{L}^{(t)}_{\text{reach}}\mathbf{J}_{S\times 1}\right\}, \tag{16}$$

the larger it is, the closer the current CON tends to be connected, and obviously be a connected CON when $\mathcal{N}_{\mathcal{S}} = S$.

E. CON ISL Resilience $\mathcal{K}_{\mathcal{S}}$ Analysis

Sequel to Section III-C, this section dedicates to *CON ISL resilience* $\mathcal{K}_{\mathcal{S}}$ (*i.e.* CON node connectivity) from ISL dimension, *i.e.* CON-centric perspective, which represents the CON ability to maintain connectivity against structural perturbations and is a key requirement for resilience CON. From a practical perspective, it quantifies how many SAT failures the CON can tolerate before the ISL network becomes disconnected. Building on node connectivity principle, we define $\mathcal{K}_{\mathcal{S}}$ to assess structural robustness against node-level failures such as SAT destruction or malfunction. Specifically, if a CON system remains a connected CON after removing $(S_{\mathcal{K}}-1)$ SATs and is a non-connected CON after removing $S_{\mathcal{K}}$ SATs, *i.e.* $\forall\mathcal{S}'\subset\mathcal{S}, |\mathcal{S}'| < S_{\mathcal{K}}, \Phi(\mathbf{L}^{(t)}_{\mathcal{S}'}) = 1$ and $\exists\mathcal{S}''\subset\mathcal{S}, |\mathcal{S}''| = S_{\mathcal{K}}, \Phi(\mathbf{L}^{(t)}_{\mathcal{S}''}) = 0$, where the connectivity determination function $\Phi(\cdot)$ is defined by eq. (17), then the CON possess node connectivity $\mathcal{K}_{\mathcal{S}}$ of strength $S_{\mathcal{K}}$. Combined with Corollary 1, $\mathcal{K}_{\mathcal{S}}^{(t)} = \max\{k\in\mathbb{N}^+ \mid \forall\mathcal{S}'\subset\mathcal{S},\ |\mathcal{S}'| < k,\ \Phi(\mathbf{L}^{(t)}_{\mathcal{S}'}) = 1\}$. $\mathbf{L}^{(t)}_{\mathcal{S}'}$ is sub-CON adjacency matrix (refer to Section II-C) after removing SAT set $\mathcal{S}'$, and $\Phi(\mathbf{L}^{(t)}_{\mathcal{S}'})$ equals

$$\exists k\in\mathbb{N}, 1\le k < |\mathcal{S}\backslash\mathcal{S}'|\,, \text{s.t.}\forall i\ne j, [(\mathbf{L}^{(t)}_{\mathcal{S}'})^k]_{ij} > 0\ ?\ 1:0, \tag{17}$$

where $\mathbf{L}^{(t)}_{\mathcal{S}'}$ can be obtained by node rejection mask matrix $\mathbf{M}_{\mathcal{S}'} \in \{0,1\}^{S\times S}$, *i.e.* $\mathbf{L}^{(t)}_{\mathcal{S}'} = \mathbf{L}^{(t)}\odot\mathbf{M}_{\mathcal{S}'}$, modeling the structural change after SAT destruction or malfunction:

$$\mathbf{M}_{\mathcal{S}'}(i,j) = 0 \text{ if } \mathfrak{s}_i\in\mathcal{S}' \text{ or } \mathfrak{s}_j\in\mathcal{S}' \text{ else } 1. \tag{18}$$

Algorithm 1 Reference-Guided Niche CON Filling $\mathfrak{N}$

Input: $\mathcal{M}_{\mathfrak{T}}$, $\mathbf{F}$, $\mathcal{F}_l$, reference point set $\mathcal{X}$.
Initialization: $\mathfrak{x}$-linked CON set $\mathcal{M}_{\mathfrak{x}} = \emptyset, \forall\mathfrak{x}\in\mathcal{X}$;
for *each* $n\in\{1,2,...,N\}$ **do**
  $\mathfrak{F}_n(\mathfrak{m}) \leftarrow \mathfrak{F}_n(\mathfrak{m}) - \min_{\mathfrak{m}\in\mathcal{M}_{\mathfrak{T}}}\mathfrak{F}_n(\mathfrak{m}), \forall\mathfrak{m}$;
  $\mathfrak{F}_n(\mathfrak{m}) \leftarrow \mathfrak{F}_n(\mathfrak{m}) / \max_{\mathfrak{m}\in\mathcal{M}_{\mathfrak{T}}}\mathfrak{F}_n(\mathfrak{m}), \forall\mathfrak{m}$; ▷ $\mathfrak{N}^{\text{norm}}$
Compute reference line $\mathfrak{x}^{\text{line}}, \forall\mathfrak{x}\in\mathcal{X}$;
for *each CON individual* $\mathfrak{m}\in\mathcal{M}_{\mathfrak{T}}$ **do**
  for *each reference point* $\mathfrak{x}\in\mathcal{X}$ **do**
    Obtain perpendicular segment $d^{\perp}(\mathfrak{m},\mathfrak{x}^{\text{line}})$;
  $\mathcal{M}_{\mathfrak{x}} = \mathcal{M}_{\mathfrak{x}}\cup\{\mathfrak{m}\} : \hat{\text{argmin}}_{\mathfrak{x}\in\mathcal{X}}\, d^{\perp}(\mathfrak{m},\mathfrak{x}^{\text{line}})$; ▷ $\mathfrak{N}^{\text{asso}}$
while $|\mathcal{M}_{\mathfrak{T}}| \le M$ **do**
  $\mathfrak{x}_{\downarrow} = \mathfrak{x} : \hat{\text{argmin}}_{\mathfrak{x}\in\mathcal{X}}|\mathcal{M}_{\mathfrak{x}}|$, temp set $\mathcal{M}^t_{\downarrow} = \emptyset$;
  for *each CON individual* $\mathfrak{m}\in\mathcal{F}_l$ **do**
    $\mathcal{M}^t_{\downarrow} = \mathcal{M}^t_{\downarrow}\cup\{\mathfrak{m}\}$ **if** $\mathfrak{x}_{\downarrow}$ in $\text{argmin}_{\mathfrak{x}\in\mathcal{X}}\, d^{\perp}(\mathfrak{m},\mathfrak{x})$;
  if $\mathcal{M}^t_{\downarrow} = \emptyset$ **then**
    Remove this reference point $\mathcal{X} = \mathcal{X}\setminus\{\mathfrak{x}_{\downarrow}\}$;
  else
    $\mathcal{M}_{\mathfrak{x}_{\downarrow}} = \mathcal{M}_{\mathfrak{x}_{\downarrow}}\cup\{\mathfrak{m}\} : \hat{\text{argmin}}_{\mathfrak{m}\in\mathcal{M}^t_{\downarrow}}\, d^{\perp}(\mathfrak{m},\mathfrak{x}_{\downarrow})$;
    $\mathcal{M}_{\mathfrak{T}} = \mathcal{M}_{\mathfrak{T}}\cup\{\mathfrak{m}\} : \hat{\underset{\mathfrak{m}\in\mathcal{M}^t_{\downarrow}}{\text{argmin}}}\, d^{\perp}(\mathfrak{m},\mathfrak{x}_{\downarrow})$; ▷ $\mathfrak{N}^{\text{nich}}$
    $\mathcal{F}_l = \mathcal{F}_l\setminus\{\mathfrak{m}\} : \hat{\text{argmin}}_{\mathfrak{m}\in\mathcal{M}^t_{\downarrow}}\, d^{\perp}(\mathfrak{m},\mathfrak{x}_{\downarrow})$.
Output: Filled parent CON population $\mathcal{M}_{\mathfrak{T}}$.

While the acquisition of $\mathbf{L}^{(t)}$ in simulation part, in conjunction with Section II-C, can directly depend on $\mathrm{L}^{(t)}_{\mathfrak{s}_i,\mathfrak{s}_j} = \left\lceil \frac{1}{2}\left\{\text{sgn}\left[||\mathbf{d}_{\mathfrak{s}_i,\mathfrak{s}_j}||^{\max}_2 - ||\mathbf{d}^{(t)}_{\mathfrak{s}_i,\mathfrak{s}_j}||_2\right] + 1\right\}\right\rceil$, where $\mathbf{d}_{\mathfrak{s}_i,\mathfrak{s}_j} = \mathbf{v}_{\mathfrak{s}_j} - \mathbf{v}_{\mathfrak{s}_i}$ is the vector from SAT $\mathfrak{s}_i$ to $\mathfrak{s}_j$. And as part of $\Phi(\cdot)$, combined with $\mathbf{L}^{(t)}$, the following Lemma associated with CON connectivity determination can be derived.

Lemma 4: If $D^{(t)}_{\mathcal{S}} < 2(S-1)$, where $D^{(t)}_{\mathcal{S}} = \sum_{s=1}^{S} D^{(t)}_{\mathfrak{s}_s} = \sum_{i=1}^{S}\left(\sum_{j=1}^{S}\mathrm{L}^{(t)}_{\mathfrak{s}_i,\mathfrak{s}_j}\right) - \text{Tr}(\mathbf{L}^{(t)})$ is the sum of CON node degrees at moment t and $\text{Tr}(\cdot)$ denotes the trace operation, the CON must be a non-connected CON at moment t.

Proof: In case of a connected CON, the minimum number of possible edges is $(S-1)$, *i.e.* $|\mathcal{L}|\ge S-1$. According to the structure of the minimum spanning tree, it is known that $D^{(t)}_{\mathcal{S}}\ge 2(S-1)$, and the contrapositive is thus proved. ∎

Then, the *CON ISL resilience* $\mathcal{K}_{\mathcal{S}}$, *i.e.* CON node connectivity, can be finally defined by the following equation ($k\in\mathbb{N}^+$):

$$\mathcal{K}_{\mathcal{S}} \triangleq \frac{1}{T}\sum_{t=1}^{T}[\max\{k \mid \forall\mathcal{S}\prime\subset\mathcal{S}, |\mathcal{S}\prime| < k, \Phi(\mathbf{L}^{(t)}_{\mathcal{S}\prime}) = 1\}]. \tag{19}$$

Although the local access resilience $\mathcal{R}^{o}_{\mathcal{G}_0}$ and the CON ISL resilience $\mathcal{K}_{\mathcal{S}}$ are mathematically defined as independent metrics from GP-centric and CON-centric perspectives, their

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coordination is realized at the offline CON design stage through joint multi-objective optimization. Rather than modeling an explicit runtime compensation mechanism, the proposed framework enhances resilience structurally: improved ISL connectivity increases path redundancy under SAT failures, which indirectly supports service persistence for covered regions, while spatial coverage redundancy also contributes to more favorable ISL topologies. Therefore, their interaction is implicitly reflected in the Pareto-optimal CON configurations, although no explicit cross-layer coupling model is introduced in this study.

IV. Design Optimization Framework

This section presents an integrated optimization framework for resilience CON design, coupling the non-dominated sorting genetic algorithm-III (NSGA-III) for CON MOP (Section IV-B) with a high-fidelity CON simulation system (Section IV-C), built upon key problem formulations (Section IV-A).

A. Remote Area-Empowered Resilience CON Formulation

Building on balanced global coverage, we focus on *communication enhancement* and *resilience reinforcement* in remote areas, navigating trade-offs shaped by both structural configurations and performance requirements. To lay the foundation for subsequent CON design in this context, we first explore configuration performance in terms of global coverage and ISL connectivity, including *i)* global access coverage performance along with its scale and hierarchical evolution; *ii)* connected CON and trade-off with global access coverage. The related formulations are described in single-objective optimization problem (SOP) (S1) as shown in eq. (20), and a further MOP (M1) additionally considers ISL connectivity.

$$(\text{S1}) \max_{h_{\mathfrak{s}},\Omega^0,M_A^0,\mathbf{I},F} \mathcal{C}_{\mathcal{G}}, \tag{20a}$$

$$\mathbf{s.t.}\quad h_{\mathfrak{s}} \le 2000, h_{\mathfrak{s}} = H_{z_2}^{z_1}, z_1, z_2 \in \mathbb{N}^+, \tag{20b}$$

$$\Omega^0 \in [0^\circ, 360^\circ],$$

$$\Omega^{s,p} \sim \underbrace{\text{eq. (2a) or (2b)}}_{\text{Walker}} \text{ or } \underbrace{(5)}_{\text{HyInc}}, \tag{20c}$$

$$M_A^0 \in [0^\circ, 360^\circ], M_A^{s,p} \sim \text{eq. (3)}, \tag{20d}$$

$$\mathbf{I} \in \underbrace{[0^\circ, 90^\circ] \cdot \mathbf{J}_{1\times P}}_{\text{Walker}} \text{ or } \mathbf{I} \in \underbrace{[0^\circ, 90^\circ]^{1\times P}}_{\text{HyInc}}, \tag{20e}$$

$$F_\omega \in \{0, 1, 2, \ldots, P-2, P-1\}, \tag{20f}$$

where eq. (20b) enables low-latency LEO CON service, *i.e.* $\Delta\tau_{\mathfrak{g}}/2 \le \Delta\tau_{\max} \xrightarrow[\theta_{\min}=10^\circ]{2000 \text{ km}} 14.8$ ms [38] according to eq. (11). Given the focus of this study, $h_{\mathfrak{s}}$ is fixed as $H_{z_2}^{z_1}$, setting the CON optimization vector $\mathbf{S} = [\Omega^0 \ M_A^0 \ \mathbf{I} \ F]$ and deferring height correlation analysis to future work. It is emphasized that like other OPs in this paper, SOP (S1) is a mixed-integer OP and all of them are *implicit* OPs. Then MOP (M1) gets

$$(\text{M1}) \max_{\mathbf{S}} \mathcal{C}_{\mathcal{G}}, \mathcal{N}_{\mathcal{S}}, \mathbf{s.t.}(20\text{b})\text{-}(20\text{e}), \underbrace{(20\text{f})}_{\text{Walker/HyInc}} \text{ or } F_\omega \in \underbrace{[0, P-1]}_{\text{HyInc-F}}, \tag{21}$$

where $F_\omega \in [0, P-1]$ indicates release of the continuous phase factor gain. In addition, as part of exploration of connected CON design, an OP study of a single-plane connected CON is considered, *i.e.* $\max_{\mathbf{s}} \mathcal{N}_{\mathcal{S}}, \mathbf{s.t.}$ constraints in (M1), mapping Lemma 5 related to CON scale lower bound of single-plane connected CON, and paves the way for subsequent MOP (M2).

Lemma 5: The CON scale lower bound for single-plane connected CON in circular CON case can be obtained from ISL LOS distance threshold $||\mathbf{d}_{\mathfrak{s}_i,\mathfrak{s}_j}||_2^{\max}$ (can short to $d_{2\mathfrak{s},\max}$):

$$S_{\min}^{\mathcal{N}}(P=1) = \left\lceil 2\pi \left\{ \arccos\left[1 - \frac{d_{2\mathfrak{s},\max}^2}{2(R_E + h_{\mathfrak{s}})^2} \right] \right\}^{-1} \right\rceil. \tag{22}$$

Proof: See Appendix D. ∎

Then, we consider local area-empowered CONs with global access coverage, with particular emphasis on *communication enhancement* and *resilience reinforcement* CON design for the empowered remote areas, including *i)* SOP (S2) as shown in eq. (23). Redundancy-based remote area enhancement CONs with balanced global access; *ii)* MOP (M2), *i.e.* integrated CONs, additionally focuses on ISL connectivity and resilience.

$$(\text{S2}) \max_{\mathbf{S}} \ \mathcal{A}_{\mathcal{G}'}, \mathbf{s.t.}\ \mathcal{C}_{\mathcal{G}} \ge \mathcal{C}_{\mathcal{G}}^{\text{threshold}}, (20\text{b})\text{-}(20\text{f}), \tag{23}$$

where $\mathcal{C}_{\mathcal{G}}^{\text{threshold}}$ is the *global access cover* threshold, serving as a tunable constraint to ensure baseline global coverage balance and enable flexible coordination of diverse coverage strategies. Based on quantitative metrics of access asset capacity and structural resilience, the remote area-oriented permanently connected integrated CON, i.e., MOP (M2), is illustrated as

$$(\text{M2}) \max_{\mathbf{S}} \mathcal{A}_{\mathcal{G}_0}, \mathcal{R}_{\mathcal{G}_0}^o, \mathcal{K}_{\mathcal{S}}, \mathbf{s.t.}\ \mathcal{N}_{\mathcal{S}} = S, (\text{M1}) \ \&\ (\text{S2}). \tag{24}$$

Under the premise of *global access cover* and *CON ISL connectivity* (permanent connected CON), MOP (M2) jointly optimizes *local access asset*, *local access resilience*, and *CON ISL resilience*, offering CON deployment guidance for remote areas in communication support of emergency, military, *etc.*

As shown in eq. (25), the multiple OPs constructed are functionally progressive and mutually supportive, revealing a logical flow from basic evaluation to remote area optimization and highlighting the synergy between *communication enhancement* and *resilience reinforcement* in CON design.

$$\begin{array}{ccc} (\text{S1}) & \xrightarrow{\text{Communication Enhancement}} & (\text{S2}) \\ \text{ISL} \downarrow & \text{Remote Area Empowerment} & \downarrow \text{ISL} \\ (\text{M1}) & \xrightarrow[\text{Resilience Reinforcement}]{} & (\text{M2}) \end{array} \tag{25}$$

And it should be noted that this study focuses on resilience-related structural metrics for pre-deployment CON design. The proposed multi-objective framework nevertheless remains extensible, and additional performance indicators, such as delay or traffic distribution metrics, can be incorporated when broader cross-layer models involving traffic demand, routing, and resource scheduling are considered.

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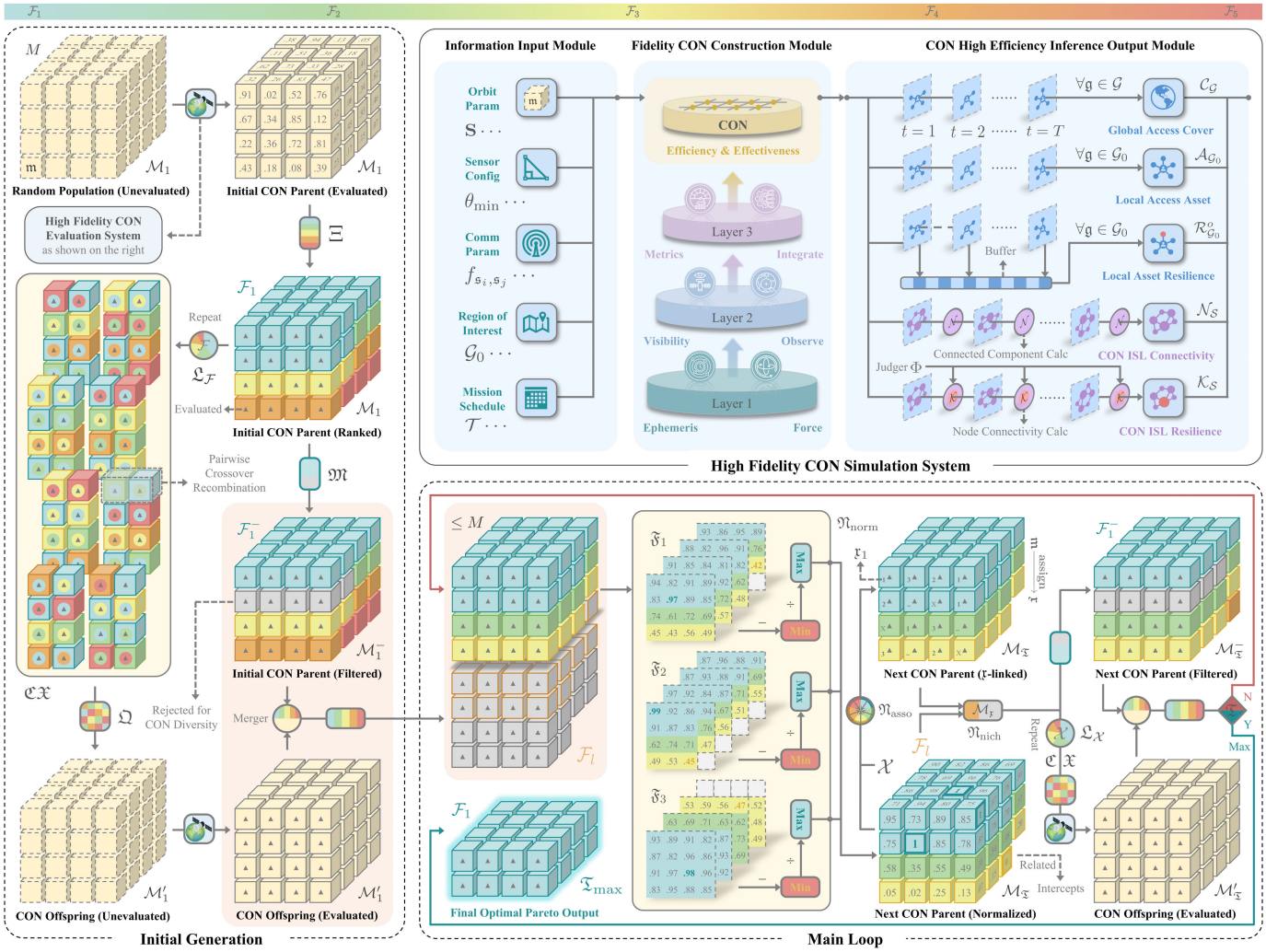


Fig. 3. Overview schematic of integrated process of the design optimization framework. (Assuming Pareto front maximum layer is 5, $N = 3$ and $\mathfrak{F}_n \in [0, 1]$).

B. NSGA-Iii-Based CON Design Framework and Algorithms

The hybrid optimization framework integrates a Pareto-based NSGA-III [39], [40] for efficient MOP solving with a high-fidelity CON simulation system for precise evaluation and closed-loop feedback. The framework overview is illustrated in Fig. 3. Compared to NSGA-II [41], NSGA-III-based CON design framework is beneficial for high-dimension diversity,[5] spatio-temporal complexity,[6] and extensibility[7]:

$$\text{NSGA-III} = (\mathcal{X}, \mathcal{M}, \mathcal{F}, \mathbf{F}, \Xi, \mathfrak{M}, \mathfrak{N}, \mathfrak{L}_{\mathcal{X}}, \mathfrak{C}, \mathfrak{X}, \mathfrak{Q}, \mathfrak{T}), \quad (26)$$

where $\mathcal{X}$, $\mathcal{M}$, and $\mathcal{F}$ are set elements associated with diversity reference point, CON population, and Pareto front, $\mathbf{F}$ is the CON integration evaluation system, and the remaining symbols refer to a series of algorithmically related operations including CON fast non-dominated sorting Ξ, $\mathfrak{M}$-fraction elitist CON truncation $\mathfrak{M}$,[8] reference point-guided niching CON filling $\mathfrak{N}$, $\mathcal{X}$-based binary tournament selection $\mathfrak{L}_{\mathcal{X}}$,[9] genetic coding $\mathfrak{C}$, crossover $\mathfrak{X}$, mutation $\mathfrak{Q}$, termination $\mathfrak{T}$.[10]

1) Normalized Plane Reference Point $\mathcal{X}$ Generation: To ensure Pareto-optimal diversity, NSGA-III employs a predefined set of reference points. These points support both automated generation under structured rules and flexible customization based on mission-specific objective distributions. Given the preference-free context of CON design, the Das and Dennis's systematic approach [42] can be applied to place $X = C^{F_X}_{N+F_X-1}$ (F_X is division factor) reference points $\mathfrak{x}_x \in \mathcal{X}$, $x = \{1, 2, \ldots, X\}$ uniformly on a normalized $(N-1)$-dimensional unit simplex with equal axis intercepts and uniform inclination across all N objectives.

2) Integrated CON Performance Evaluation System $\mathbf{F}$: The CON evaluation $\mathbf{F}(\mathfrak{m}) = [\mathrm{F}_1(\mathfrak{m})\ \mathrm{F}_2(\mathfrak{m})\ \cdots \mathrm{F}_\mathrm{N}(\mathfrak{m})]$ directly depends on the objective function $\mathfrak{F}(\mathfrak{m})$ for performance

[5]NSAG-II-like algorithm's crowdedness mechanism fails in high dimension space due to the exponential growth in non-dominated solution and indistinguishable Euclidean distances, limiting Pareto front convergence and diversity.

[6]Larger computational and storage overheads exist when realizing population diversity using methods such as NSGA-II crowdedness or clustering.

[7]NSGA-III simplifies the mapping between individual and diversity through a discrete reference point mechanism, making its complexity growth smoother and more suitable for later expansion of high-dimension CON design.

[8]For CON diversity, limits top non-dominated parent merge under elitism.

[9]Repeatedly sample two CON individuals with replacement, compare via reference set $\mathcal{X}$, and retain the better one until the population reaches M.

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Algorithm 2 CON Fast Non-Dominated Sorting Ξ

Input: All CON individual $\mathfrak{m} \in \mathcal{M}$, evaluator $\mathbf{F}$.
Initialization: Pareto front counter $l = 1$;
for *each CON individual* $\mathfrak{m} \in \mathcal{M}$ **do**
  $\mathcal{D}^{\mathfrak{m}} = \emptyset$, $|\mathcal{D}_{\mathfrak{m}}| = 0$;
  for *other CON individual* $\mathfrak{m}' \in \mathcal{M}$ $(\leftarrow \mathbf{F})$ **do**
    $\mathcal{D}^{\mathfrak{m}} = \mathcal{D}^{\mathfrak{m}} \cup \{\mathfrak{m}'\}$ **if** $\mathfrak{m} \succ \mathfrak{m}'$ **else** $|\mathcal{D}_{\mathfrak{m}}|$++;
  1st Pareto front $\mathcal{F}_1 = \mathcal{F}_1 \cup \{\mathfrak{m}\}$ **if** $|\mathcal{D}_{\mathfrak{m}}| = 0$;
while l^{th} *Pareto front* $\mathcal{F}_l \neq \emptyset$ **do**
  Temporary set $\mathcal{F}_{\text{temp}} = \emptyset$ for next Pareto front;
  for *each CON individual* $\mathfrak{m} \in \mathcal{M}$ **do**
    for *each CON individual* $\mathfrak{m}' \in \mathcal{D}^{\mathfrak{m}}$ **do**
      $|\mathcal{D}_{\mathfrak{m}'}| = |\mathcal{D}_{\mathfrak{m}'}| - 1$;
      $\mathcal{F}_{\text{temp}} = \mathcal{F}_{\text{temp}} \cup \{\mathfrak{m}'\}$ **if** $|\mathcal{D}_{\mathfrak{m}'}| = 0$;
  $l = l + 1$, $\mathcal{F}_l = \mathcal{F}_{\text{temp}}$.
Output: $\mathcal{D}^{\mathfrak{m}}, |\mathcal{D}_{\mathfrak{m}}|, \forall \mathfrak{m} \in \mathcal{M}$; $\mathcal{F}_l, \forall l$.

Algorithm 3 $\mathcal{X}$-Based NSGA-III (Main Loop)

Input: $\mathfrak{M}$-filtered parent CON $\mathcal{M}_1^-$, evaluator $\mathbf{F}$, offspring CON $\mathcal{M}_1'$, reference point set $\mathcal{X}$.
Initialization: Current generation $\mathfrak{T} = 2$;
while *current generation* $\mathfrak{T} \leq \mathfrak{T}_{\max}$ **do**
  Front $\mathcal{F}_l, \forall l \leftarrow \Xi(\mathcal{M}_{\mathfrak{T}-1}^- \cup \mathcal{M}_{\mathfrak{T}-1}', \mathbf{F})$; ▷ Alg. 2
  $\mathcal{M}_{\mathfrak{T}} = \emptyset$ for next parent CON, $l = 1$;
  while $|\mathcal{M}_{\mathfrak{T}}| + |\mathcal{F}_l| \leq M$ **do**
    $\mathcal{M}_{\mathfrak{T}} = \mathcal{M}_{\mathfrak{T}} \cup \mathcal{F}_l$, $l = l + 1$;
  Niche filling $\mathcal{M}_{\mathfrak{T}} \leftarrow \mathfrak{N}(\mathcal{M}_{\mathfrak{T}}, \mathbf{F}, \mathcal{F}_l, \mathcal{X})$; ▷ Alg. 1
  $\mathcal{F}_1^- = \mathcal{F}_1[1 : M\mathfrak{M}]$, $\mathcal{M}_{\mathfrak{T}}^- = \mathcal{M}_{\mathfrak{T}} \setminus (\mathcal{F}_1 \setminus \mathcal{F}_1^-)$;
  $\mathcal{M}_{\mathfrak{T}}' \xleftarrow{\mathbf{F},\mathfrak{Q},\mathfrak{X},\mathfrak{C},\mathfrak{L}_{\mathcal{X}}} \mathcal{M}_{\mathfrak{T}}$, $\mathfrak{T} = \mathfrak{T} + 1$.
Output: First front-related $\mathcal{F}_1^-$ in generation $\mathfrak{T}_{\max}$.

assessment of CON $\mathfrak{m}_m \in \mathcal{M}, m = 1, 2, \ldots, M$ as follows:

$$[\mathbf{F}(\mathfrak{m})]^{\mathrm{T}} = \begin{bmatrix} (\mathfrak{F}_1(\mathfrak{m}) + C_{\downarrow}^{+} > 0) \ ? \ (\mathfrak{F}_1(\mathfrak{m}) + C_{\downarrow}^{+}) : 0 \\ \vdots \\ (\mathfrak{F}_N(\mathfrak{m}) + C_{\downarrow}^{+} > 0) \ ? \ (\mathfrak{F}_N(\mathfrak{m}) + C_{\downarrow}^{+}) : 0 \end{bmatrix}, \tag{27}$$

where $C_{\downarrow}^{+}$ is a suitably small positive constant, and M represents the CON population size. As the most time-consuming and complex part of the entire optimization process, the evaluation of $\mathfrak{F}(\mathfrak{m})$ relies on a high-fidelity CON performance assessment mechanism, integrating orbit modeling, visibility calculation, and constraint evaluation, details in Section IV-C.

3) CON Fast Non-Dominated Sorting Operation Ξ: The process shown in Algorithm 2 performs Pareto front sorting for all CON individuals based on their fitness values, with inputs including all CONs $\mathfrak{m} \in \mathcal{M}$ and the evaluation system $\mathbf{F}$. $\mathcal{D}^{\mathfrak{m}}$ and $\mathcal{D}_{\mathfrak{m}}$ represent the sets of CONs dominated by and dominating CON $\mathfrak{m}$, respectively, where $\mathfrak{m} \succ \mathfrak{m}'$ indicates that CON $\mathfrak{m}$ dominates CON $\mathfrak{m}'$, *i.e.* $\mathrm{F}_n(\mathfrak{m}) > \mathrm{F}_n(\mathfrak{m}'), \forall n \in \{1, 2, \ldots, N\}$. $\leftarrow \mathbf{F}$ indicates that the relevant step involves CON evaluation system. The output of this operation can include $\mathcal{D}^{\mathfrak{m}}, |\mathcal{D}_{\mathfrak{m}}|, \forall \mathfrak{m} \in \mathcal{M}$ and $\mathcal{F}_l, \forall l$, where $\mathcal{F}_l$ represents the set of CONs in the lth Pareto front.

[10] $\mathfrak{C}$, $\mathfrak{X}$, and $\mathfrak{Q}$ follow basic genetic algorithm. Termination is based on a typical setting, *i.e.* the process ends when generation $\mathfrak{T}$ reaches $\mathfrak{T}_{\max}$.

4) Reference Point-Guided Niching CON Filling $\mathfrak{N}$: Instead of crowdedness, normalized reference point mechanism is introduced in high-dimension objective space [43]. As shown in Algorithm 1, this operation involves three key stages, *i.e.* normalization $\mathfrak{N}^{\text{norm}}$, association $\mathfrak{N}^{\text{asso}}$, and niching $\mathfrak{N}^{\text{nich}}$. *First*, axis-zeroing (Step 3) and normalization (Step 4) are applied. *Second*, the reference lines $\mathfrak{x}^{\text{line}}, \forall \mathfrak{x} \in \mathcal{X}$, representing the straight lines between the reference points and the origin, are calculated, and the CON individuals are associated with corresponding reference points based on perpendicular segment rule (Steps 6-9). *Third*, following the minimum associated reference point priority rule, parent CON population undergoes niche filling until reaches M (Steps 10-19). $\hat{\arg\min}(\cdot)$ denotes an extraction operator that returns a randomly selected element when multiple minima exist.

C. Fidelity-Aware Integrated Performance Evaluation Module

Modular CON simulation system solves implicit objectives and seamlessly integrates with NSGA-III, as shown in Fig. 3. The external optimizer and internal simulator interact via standardized interfaces, supporting reconfigurable modeling for diverse mission and CON configurations. The following core layers underpin the fidelity-aware CON construction, enabling efficient and accurate performance inference.

1) Ephemeris and Force Modeling Layer: To support precise orbit-level evaluation of multi-dimension metrics, a task-adaptive dynamics environment is established, featuring modular integration of *i)* perturbation-aware propagators from baseline TwoBody to extended J2/J4 models; *ii)* WGS-84-based georeferencing with atmospheric density approximation via the 1976 US Standard Atmosphere; *iii)* unified time and attitude reference systems supporting multiple time-scale mappings and long-term frame modeling.

2) Visibility and Observation Modeling Layer: To empower communication and resilience in remote area scenarios, a visibility analysis module is layered atop the orbit engine, enabling unified ST and ISL LOS modeling. Built on WGS-84 with configurable elevation and dwell constraints, it employs high-frequency temporal sweeping to generate dynamic service windows. Its decoupled design supports scalable scheduling in multi-area and high-concurrency missions.

3) Performance Metric Integration Layer: This layer integrates core performance metrics including coverage, connectivity, and resilience, enabling efficient CON evaluation. It relies on trusted data such as SAT epimerizes and metrics are calculated through visibility and connectivity simulations, with results aggregated across time and areas. Evaluations feedback to external optimizer via standardized interfaces, ensuring a fast and stable feedback loop.

Modular decoupling enables future integration of advanced algorithms, surrogate models, or cross-mission adaptation.

V. Simulation Results

Grounded in the theoretical constructs of Section II and Section III, and the integrated CON formulations articulated in Section IV, this section presents and analyzes the combined simulation results. Table I shows essential settings [44].

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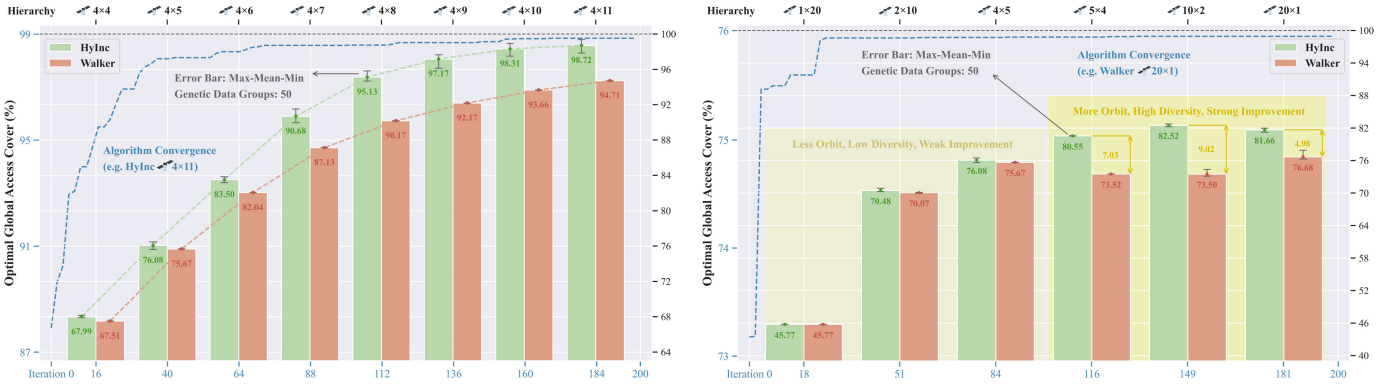


Fig. 4. (S1) related. (a) Evolutionary trend comparison of global access cover expansion; (b) Impact comparison of CON hierarchy on global access cover.

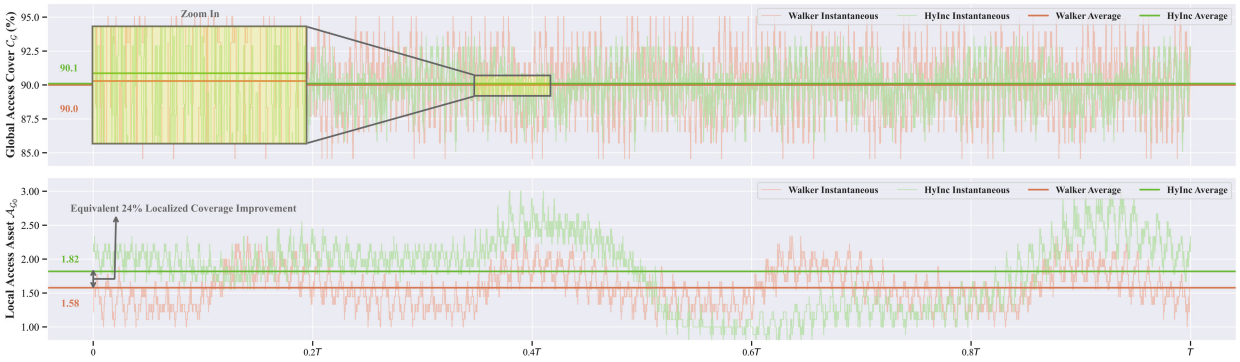


Fig. 5. Temporal performance of communication enhancement in remote areas considering global access coverage. ($\mathcal{C}_{\mathcal{G}}^{\text{threshold}} = 0.9$, hierarchy: 4×8).

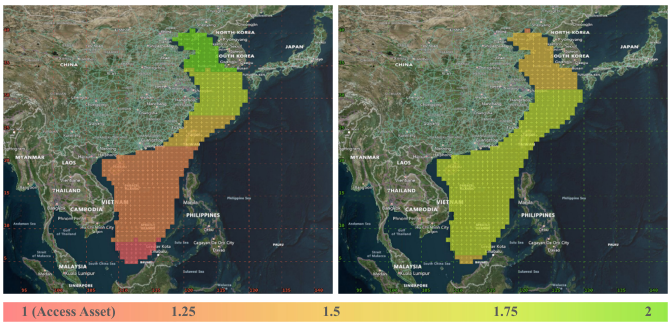


Fig. 6. Heatmap of local access asset performance. (←: Walker; →:HyInc).

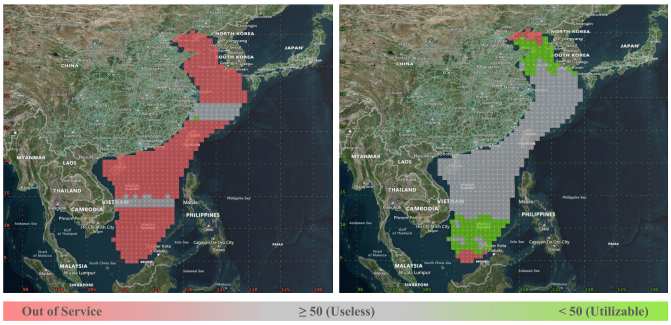


Fig. 7. Heatmap of local GDOP performance. (←: Walker; →:HyInc).

TABLE I

MAIN SETTINGS OF THE FRAMEWORK FOR SIMULATION

Parameter	Value
Gravity model, T(sec), Time step(sec)	GL0660B, 86164, 1
Propagator, Gravitation(km^3/s^2)	TwoBody, 4902.8
Crossover fraction, Mutation rate	0.9, 0.005
R_E(km), $\theta_{\min}(^\circ)$, M, $\mathfrak{M}$, $\mathfrak{T}_{\max}$	6378.14, 10, 100, 0.3, 200

A. Performance Analysis for Global Access Cover

To preliminarily validate HyInc's structural advantages and performance potential, Fig. 4(a) and Fig. 4(b) illustrate simulation results of SOP (S1) under two distinct configuration strategies. The evaluation targets *global access cover*, with multiple independent runs ensuring statistical robustness, shown via *max-mean-min* error bar. As shown by the blue dashed lines, taking hierarchies "4×11" in HyInc and "20×1" in Walker as exemplars, the convergence profiles exhibit strong stability and indicates both search accessibility and solution robustness under the current optimization scheme. It is worth noting that the CON scales considered here correspond to a structurally representative design regime under high-fidelity simulation-based evaluation. Since each fitness assessment relies on performance computation, the adopted scale ensures a balance between statistical robustness and computational feasibility, while still sufficient to reveal performance trends.

For fixed $P = 4$, Fig. 4(a) compares the evolution of *global access cover* as S^P increases. Overall, HyInc consistently surpasses Walker, with advantages amplifying at medium-to-large scales. For instance, at hierarchy "4×8", HyInc achieves 95.13% vs. Walker's 90.17%, yielding ∼5% gain, and reaches 97.17% with a 5% advantage at hierarchy "4×11". Further analysis reveals that as CON scales up, Walker exhibits diminishing marginal coverage gains and early performance saturation. In contrast, HyInc leverages inclination tuning to enhance spatial uniformity and structural diversity, effectively

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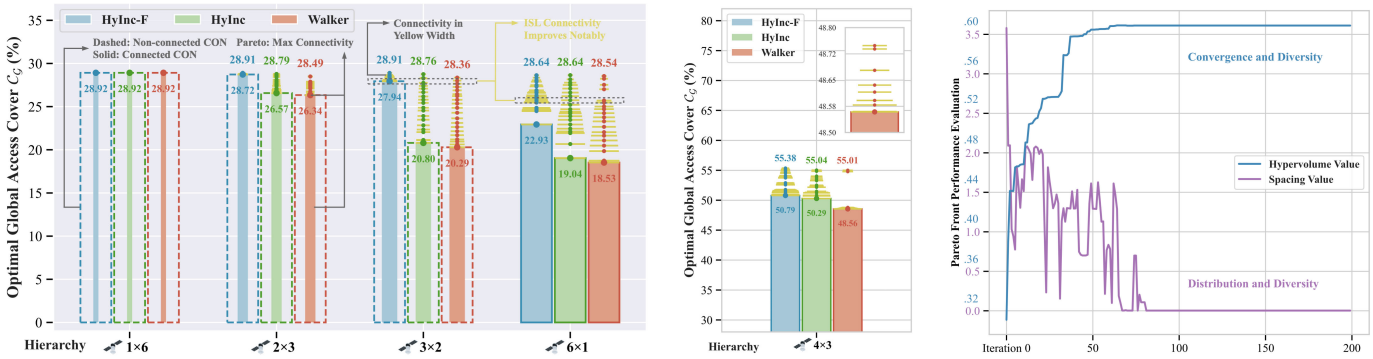


Fig. 8. (M1) related. ($C_{\mathcal{G}}^{\text{threshold}} = 0.9$) (a) Pareto space analysis on different hierarchies; (b) Inflection insight; (C) Convergence and related.

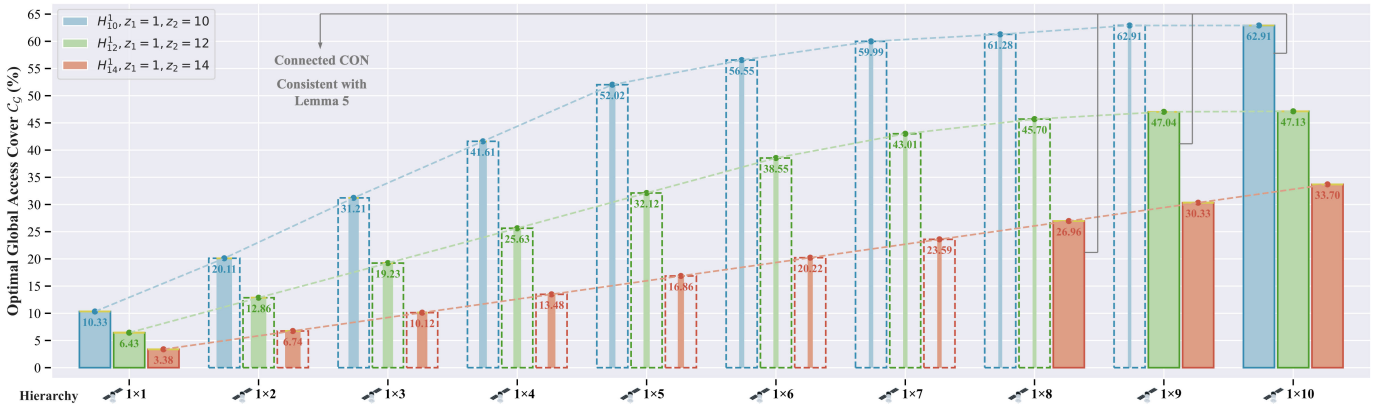


Fig. 9. Threshold scales for single-orbit plane CON connectivity for different orbital altitude scenarios. ($C_{\mathcal{G}}^{\text{threshold}} = 0.9$).

mitigating redundancy in large-scale deployment and demonstrating superior extensibility and configurability.

Fig. 4(b) compares *global access cover* under fixed $S = 20$ across varying hierarchies, highlighting how structural choices affect performance under constrained resources and revealing configurational flexibility differences. Results demonstrate HyInc's superior coverage, especially with more orbit planes, achieving 7.03% gain at "5 × 4" and 9.02% at "10 × 2", attributed to its inclination diversity, which enhances distribution flexibility and service range in multi-plane layouts.

Overall, results confirm HyInc's structural adaptability and performance advantage, especially in multi-plane scenarios, where diversity boosts service capability without adding SATs.

B. Asset Optimization for Remote Area Enhancement

Based on *global access cover* demands, this section further investigates SOP (S2), focusing on communication enhancement for remote areas, *e.g.* Chinese territory marine.

Fig. 5 illustrates the communication enhancement performance considering *global access cover*. A quantitative comparison is provided between HyInc and Walker under SOP (S2), with thin and thick lines denoting instantaneous and average values, respectively. Using Chinese territorial marines as a case study, HyInc achieves a 0.24 gain in *local access asset*, equivalent to 24% redundant access enhancement, significantly outperforming Walker and also underscoring the importance of redundancy in enhancing communication in remote areas.

Fig. 6 and Fig. 7 present heatmap of *local access asset* and geometric dilution of precision (GDOP) (reflect spatial geometry quality of observable SATs) from the CON simulation system, where demonstrate that HyInc improves both overall access assets and local coverage balance, while also enhancing the spatial geometric distribution for stronger regional support.

C. Joint ISL Connectivity and Global Coverage Assessment

As noted in MOP (M1), this section optimizes CON connectivity structure, aiming to jointly enhance *global access cover* and *CON ISL connectivity*, which is a key step from basic coverage (S1) to structural enhancement. Fig. 8(a) shows Pareto results for small-scale scenarios and the structural connectivity superiority of the proposed configuration paradigm over the classical Walker baseline under comparable orbital resource constraints, including the significant enhancement of ISL connectivity. This stems from continuous phase factor while preserving inclination diversity.

Each scatter with its horizontal line denotes a Pareto solution, where line length indicates $\mathcal{N}_{\mathcal{S}}$ and longer implies stronger connectivity. In particular, the bar highlights the Pareto solution with optimal $\mathcal{N}_{\mathcal{S}}$, and the widest bar denotes a connected CON. It reflects *global access cover* under prioritized $\mathcal{N}_{\mathcal{S}}$, where higher points imply greater trade-offs. This visualization enhances decision guidance and highlights nonlinear shifts, *e.g.* Walker in Fig. 8(b), early gains come at steep $\mathcal{N}_{\mathcal{S}}$ costs, followed by a mutation point with rapid improvement, offering a clear engineering decision anchor.

The hypervolume and spacing metrics in Fig. 8(c) validate the optimization framework's convergence and solution quality in the MOP scenario. It is worth noting that the constellation optimization considered in this work corresponds to an offline structural design stage prior to deployment. Therefore, the

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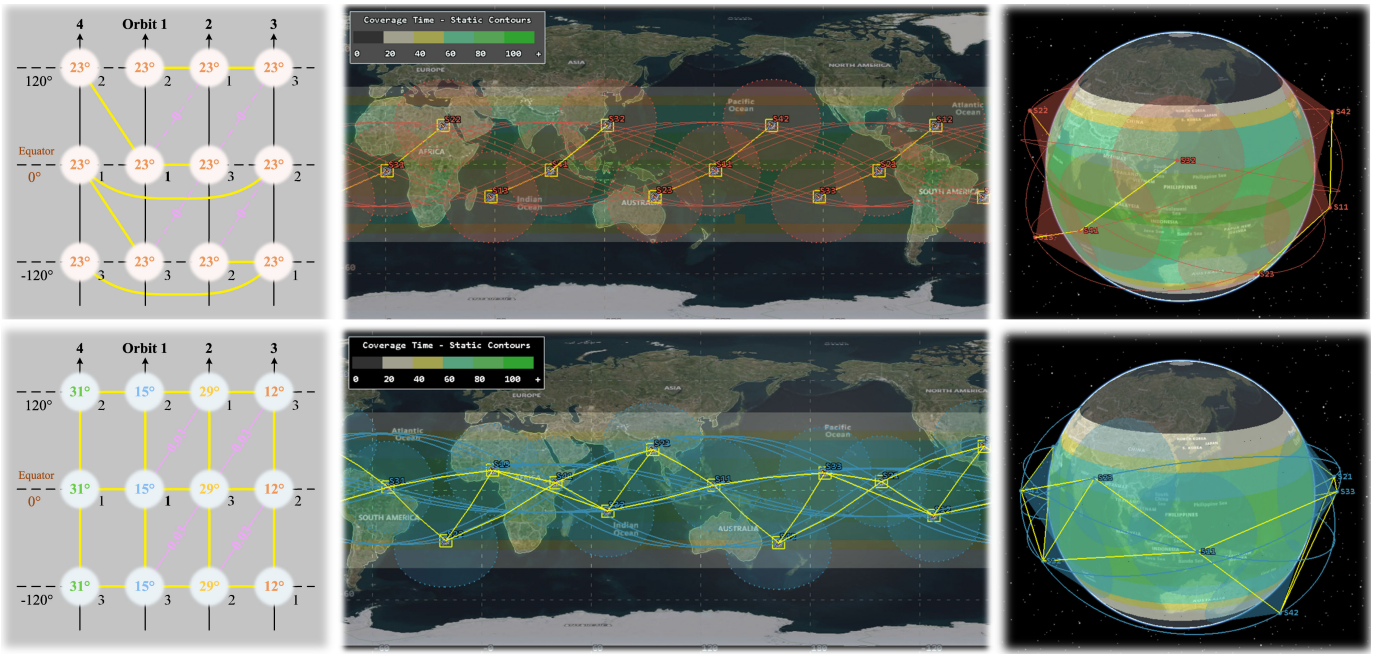


Fig. 10. Configuration comparison of a hierarchy time-slice for connectivity evaluation. ($\mathcal{C}_{\mathcal{G}}^{\text{threshold}} = 0.9$) (a) Top: HyInc-F; (b) Bottom: Walker.

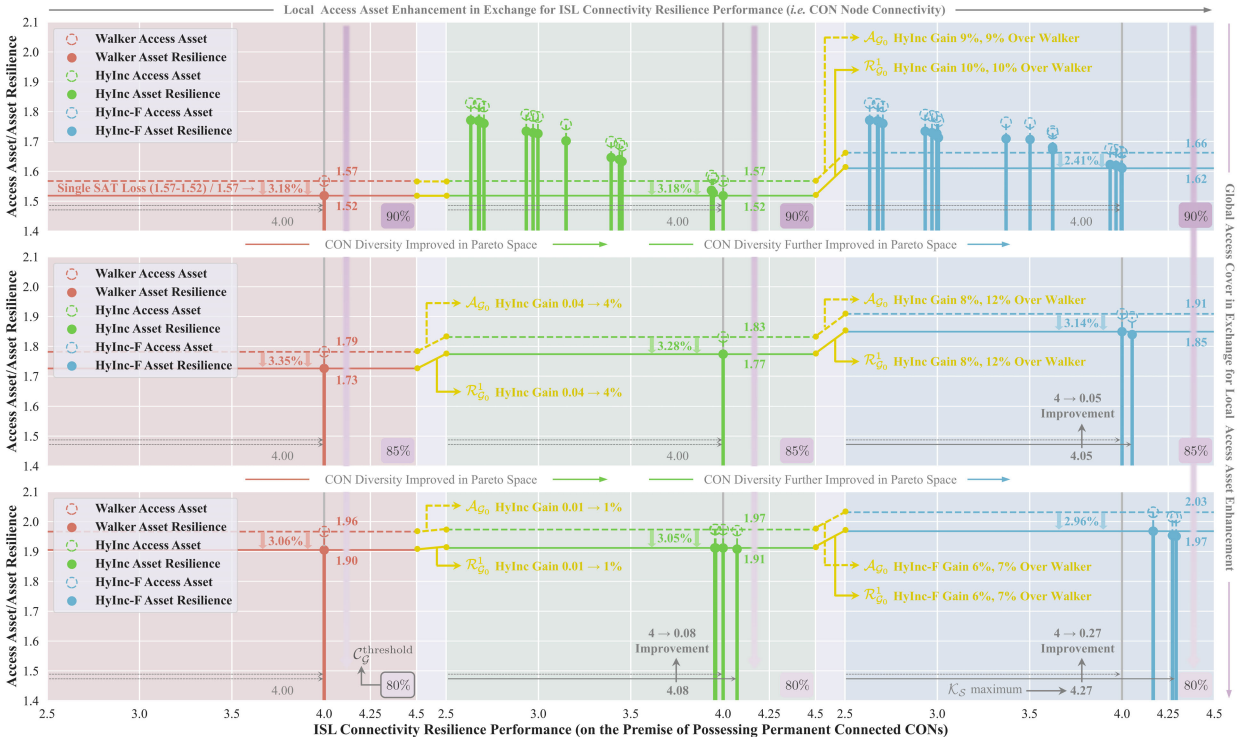


Fig. 11. Integrated CON design and configuration comparison for focused communication enhancement and resilience reinforcement. (Hierarchy: 4×8).

primary focus lies on convergence behavior and solution quality rather than strict runtime profiling. From an engineering perspective, robust Pareto convergence is more critical for pre-deployment constellation architecture determination. Fig. 9 is related to the previously described single-plane connected CON. By increasing CON scale in a single plane, $\mathcal{N}_{\mathcal{S}}$ remains constrained until a critical threshold is reached, beyond which full connectivity rapidly emerges ($\mathcal{N}_{\mathcal{S}} = S$), aligning with Lemma 5. In addition, Fig. 10 complements Fig. 8(b), offering time-slice visualizations of $\mathcal{C}_{\mathcal{G}}$ and $\mathcal{N}_{\mathcal{S}}$ for HyInc-F and Walker to aid intuitive understanding.

D. Integrated CON Optimization and Trade-off Analysis

Fig. 11 illustrates MOP (M2) design for permanent connected resilience CON, balancing global coverage and local enhancement, and analyzing trade-offs and performance shaped by structural and functional demands.

The yellow-highlighted region in Fig. 11 highlights HyInc superiority, especially the extended HyInc-F with continuous phase gain. It boosts CON diversity and overall performance, *e.g.* at an 85% $\mathcal{C}_{\mathcal{G}}^{\text{threshold}}$, HyInc surpasses Walker by 0.04 in both $\mathcal{A}_{\mathcal{G}_0}$ and $\mathcal{R}_{\mathcal{G}_0}^1$, equivalent to 179% baseline coverage plus

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$$\mathcal{R}^o_{\mathcal{G}_0} = \frac{1}{C^o_S}\sum_{k=1}^{C^o_S}\mathcal{A}^{\mathcal{S}^o_k}_{\mathcal{G}_0} = \frac{1}{C^o_S G_0 T}\sum_{t=1}^{T}\left(\left\langle \mathbf{B}^{(t)}_{\mathcal{S}^o_1}, \mathbf{J}_{G_0\times S}\right\rangle + \left\langle \mathbf{B}^{(t)}_{\mathcal{S}^o_2}, \mathbf{J}_{G_0\times S}\right\rangle + \left\langle \mathbf{B}^{(t)}_{\mathcal{S}^o_3}, \mathbf{J}_{G_0\times S}\right\rangle + \cdots + \left\langle \mathbf{B}^{(t)}_{\mathcal{S}^o_{C^o_S}}, \mathbf{J}_{G_0\times S}\right\rangle\right)$$

$$= \frac{1}{C^o_S G_0 T}\sum_{t=1}^{T}\left\{\left[C^o_S - \underbrace{\left(\overbrace{\frac{(C^o_S\cdot o)}{S}}^{\text{Masked Column Index Buffer}}\right)}_{\text{Reshape into Multiple } \mathbf{B}^{(t)}}\right]\left\langle \mathbf{B}^{(t)}, \mathbf{J}_{G_0\times S}\right\rangle\right\} \Leftrightarrow \frac{1}{C^o_S G_0 T}\sum_{t=1}^{T}\left\{\left[\underbrace{\left(\overbrace{\frac{(C^o_S\cdot (S-o))}{S}}^{\text{Retained Column Index Buffer}}\right)}_{\text{Reshape into Multiple } \mathbf{B}^{(t)}}\right]\left\langle \mathbf{B}^{(t)}, \mathbf{J}_{G_0\times S}\right\rangle\right\}$$

$$= \frac{1}{C^o_S T G_0}\sum_{t=1}^{T}\left[C^o_S\left(1-\frac{o}{S}\right)\left\langle \mathbf{B}^{(t)}, \mathbf{J}_{G_0\times S}\right\rangle\right] = \frac{S-o}{STG_0}\sum_{t=1}^{T}\left\langle \mathbf{B}^{(t)}, \mathbf{J}_{G_0\times S}\right\rangle = \frac{S-o}{STG_0}\left\langle \mathbf{B}, \mathbf{J}_{G_0\times TS}\right\rangle \tag{28}$$

4% gain. While HyInc-F further raises to 1.91 (191%), marking an 8% improvement over HyInc and 12% over Walker. Notably, these gains are achieved under guaranteed *global access cover* $\mathcal{C}_{\mathcal{G}}$ and *CON ISL connectivity* $\mathcal{N}_{\mathcal{S}}$ (permanent connected CON), while also enhancing *CON ISL resilience* $\mathcal{R}_{\mathcal{S}}$ and mitigating local access loss from SAT failures to bolster remote network stability.

The purple-highlighted region in Fig. 11 quantifies a key trade-off, *i.e.* moderately relaxing $\mathcal{C}_{\mathcal{G}}$ (*e.g.* from 90% to 85% and 80% as shown) yields notable gains in local access and resilience. Taking the right-side HyInc-F as an example, relaxing $\mathcal{C}_{\mathcal{G}}$ from 90% to 85% yields $0.25 \rightarrow 25\%$ and $0.23 \rightarrow 23\%$ gains in $\mathcal{A}_{\mathcal{G}_0}$ and $\mathcal{R}^1_{\mathcal{G}_0}$, respectively. Yet further easing to 80% brings smaller increments (14% $\mathcal{A}_{\mathcal{G}_0}$ and 12% $\mathcal{R}^1_{\mathcal{G}_0}$), revealing diminishing returns and offering practical trade-off insight for resource allocation in remote area communication.

Overall, by leveraging orbital independence and continuous phase offset, HyInc-F effectively breaks the configuration bottleneck in multi-dimension co-optimization for remote area *communication enhancement* and *resilience reinforcement*.

VI. Conclusion

This work has presented a unified, extensible framework for resilience LEO CON design for remote areas. The proposed HyInc/HyInc-F expands design space with scalable, inclination-adaptive constructs, offering superior access and resilience. The evaluation perspectives and hybrid optimization framework involved are effective and of referential value. Beyond technical innovations, this study offers transferable methodologies and strategic tools for future CON plans. Future extensions may further integrate HyInc-series structural principles with cross-orbit deployment, protocol-level co-design, and large-scale on-orbit reconfiguration under practical constraints. Surrogate-assisted rapid evaluation and computational-cost profiling are also promising directions for extending this offline framework toward real-time space digital twin-enabled CON assessment, where performance gains, efficiency, and system-level trade-offs can be jointly investigated.

Appendix A
Proof of Lemma 1

The kth power $\mathbf{L}^k$ of $\mathbf{L}$, $(\mathbf{L}^k)_{ij}$ can be derived by

$$\rightarrow \sum_{r_1=1}^{S}\sum_{r_2=1}^{S}\cdots\sum_{r_{k-1}=1}^{S}\prod_{m=0}^{k-1}\mathrm{L}_{r_m r_{m+1}}, \text{ where } r_0 = i, r_k = j$$

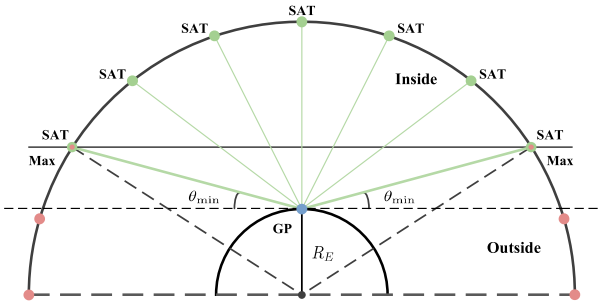


Fig. 12. Simple geometric schematic of the ST link availability.

$$\rightarrow \sum_{r_1=1}^{S}\sum_{r_2=1}^{S}\cdots\sum_{r_{k-1}=1}^{S}\mathrm{L}_{ir_1}\mathrm{L}_{r_1r_2}\cdots\mathrm{L}_{r_{k-2}r_{k-1}}\mathrm{L}_{r_{k-1}r_j}$$

$$\rightarrow \sum_{r_1=1}^{S}\mathrm{L}_{ir_1}\sum_{r_2=1}^{S}\mathrm{L}_{r_1r_2}\sum_{r_3=1}^{S}\cdots\sum_{r_{k-1}=1}^{S}\mathrm{L}_{r_{k-2}r_{k-1}}\mathrm{L}_{r_{k-1}j},$$

where $\prod$ denotes product operation. The equation counts path number $|\mathcal{Q}^k(\mathfrak{s}_i, \mathfrak{s}_j)|$ from SAT $\mathfrak{s}_i$ to $\mathfrak{s}_j$, and self-path set to 1.

Taking $S = 3$ for example, $(\mathbf{L}^2)_{ij} = \sum_{r=1}^{S}\mathrm{L}_{ir}\mathrm{L}_{rj}$ enumerates all intermediate nodes from SAT $\mathfrak{s}_i$ to $\mathfrak{s}_j$ and observes whether $i \rightarrow r$ and $r \rightarrow j$ are both connected, and plus 1 if true. Specifically, take the following $\mathbf{L}$ as an example:

$$\mathbf{L} = \begin{bmatrix}1&0&1\\0&1&1\\1&1&1\end{bmatrix} \Rightarrow (\mathbf{L}^2)_{12} = \overbrace{\mathrm{L}_{11}\mathrm{L}_{12}}^{0} + \overbrace{\mathrm{L}_{12}\mathrm{L}_{22}}^{0} + \overbrace{\mathrm{L}_{13}\mathrm{L}_{32}}^{1},$$

i.e., it indicates that there is a path of length 2 from SAT $\mathfrak{s}_1$ to $\mathfrak{s}_2$, with the path being $\mathfrak{s}_1 \rightarrow \mathfrak{s}_3 \rightarrow \mathfrak{s}_2$. Thus, if there exists a positive integer $k < S$ such that $(\mathbf{L}^k)_{ij} > 0$, then it means that there exists at least one path of length k from SAT node $\mathfrak{s}_i$ to $\mathfrak{s}_j$, and thus SAT nodes $\mathfrak{s}_i$ and $\mathfrak{s}_j$ are connected.

Appendix B
Proof of Lemma 2

Vectors $\mathbf{v}^{(t)}_{\mathfrak{g}} = [x^{(t)}_{\mathfrak{g}}\ y^{(t)}_{\mathfrak{g}}\ z^{(t)}_{\mathfrak{g}}]$ and $\mathbf{v}^{(t)}_{\mathfrak{s}} = [x^{(t)}_{\mathfrak{s}}\ y^{(t)}_{\mathfrak{s}}\ z^{(t)}_{\mathfrak{s}}]$ are the WGS-84 coordinates of GP $\mathfrak{g}$ and SAT $\mathfrak{s}$ at moment t, which satisfies $r_{\mathfrak{s}} = ||\mathbf{v}^{(t)}_{\mathfrak{s}}||_2 = R_E + h_{\mathfrak{s}} = ||\mathbf{v}^{(t)}_{\mathfrak{g}}||_2 + h_{\mathfrak{s}}, \forall t \in \mathcal{T}$. From Fig. 12, it can be seen that when SAT s is outside the range of the minimum elevation angle $\theta_{\min}$ of GP g, the magnitude of the distance vector between both is larger than when it is within the range, *i.e.* $||\mathbf{d}^{(t)}_{\mathfrak{g},\mathfrak{s}}||^{\text{outscope}}_2 > ||\mathbf{d}^{(t)}_{\mathfrak{g},\mathfrak{s}}||^{\text{in-scope}}_2$, at which point it is recognized that the GP $\mathfrak{g}$ is not within CON access range.

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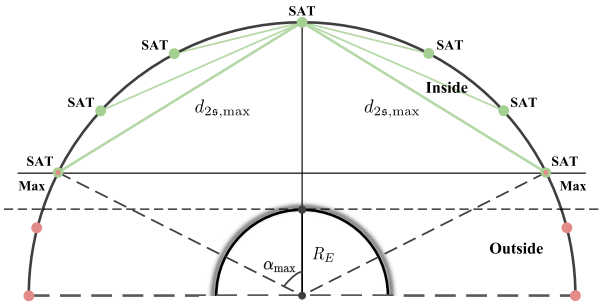


Fig. 13. Simple geometric schematic of the ISL link availability.

Thus according to the triangular cosine theorem $||\mathbf{v}_{\mathfrak{s}}^{(t)}||_2^2 = ||\mathbf{v}_{\mathfrak{g}}^{(t)}||_2^2 + ||\mathbf{d}_{\mathfrak{g},\mathfrak{s}}^{(t)}||_2^2 - 2||\mathbf{v}_{\mathfrak{g}}^{(t)}||_2 \cdot ||\mathbf{d}_{\mathfrak{g},\mathfrak{s}}^{(t)}||_2 \cdot \cos\left(\frac{\pi}{2}+\theta\right)$, the LOS distance threshold $||\mathbf{d}_{\mathfrak{g},\mathfrak{s}}||_2^{\max} = \frac{1}{2}\{-2||\mathbf{v}_{\mathfrak{g}}||_2 \sin\theta_{\min} \pm \sqrt{[2||\mathbf{v}_{\mathfrak{g}}||_2\sin\theta_{\min}]^2 - 4(||\mathbf{v}_{\mathfrak{g}}||_2^2 - ||\mathbf{v}_{\mathfrak{s}}||_2^2)}\} = -C_E \pm \sqrt{C_E^2 + h_{\mathfrak{s}}^2 + 2R_E h_{\mathfrak{s}}} \xrightarrow{\sin\theta_{\min}\geq 0} \sqrt{C_E^2 + h_{\mathfrak{s}}^2 + 2R_E h_{\mathfrak{s}}} - C_E$.

APPENDIX C
PROOF OF LEMMA 3

According to eq. (13) the number of available SATs above GP $\mathfrak{g}_g$ at moment t is $S_{\mathfrak{g}_g}^{(t)} = \sum_{s=1}^{S} b_{\mathfrak{g}_g,\mathfrak{s}_s}^{(t)}$, *i.e.*, $S_{\mathfrak{g}_g}^{(t)} = \mathbf{B}_{g,:}^{(t)}\mathbf{J}_{S\times 1}, \forall t \in \mathcal{T}, \forall g = \{1,2,\ldots,G_0\}$, where $\mathbf{B}_{g,:}$ denotes the gth row vector of matrix $\mathbf{B}$. Then $\mathcal{A}_{\mathfrak{g}_g}$ can be obtained by $\mathcal{A}_{\mathfrak{g}_g} = \frac{1}{T}\sum_{t=1}^{T} S_{\mathfrak{g}_g}^{(t)} = \frac{1}{T}\sum_{t=1}^{T}\mathbf{B}_{g,:}^{(t)}\mathbf{J}_{S\times 1}$, $\forall g = \{1,2,\ldots,G_0\}$, and $\mathcal{A}_{\mathcal{G}_0}$ exists a derivation as follows:

$$\mathcal{A}_{\mathcal{G}_0} = \frac{1}{G_0 T}\sum_{t=1}^{T}(\mathbf{B}_{1,:}^{(t)}\mathbf{J}_{S\times 1} + \mathbf{B}_{2,:}^{(t)}\mathbf{J}_{S\times 1} + \cdots + \mathbf{B}_{G_0,:}^{(t)}\mathbf{J}_{S\times 1})$$

$$= \frac{1}{G_0 T}\sum_{t=1}^{T}[(\mathbf{B}^{(t)}\mathbf{J}_{S\times 1})\mathbf{J}_{1\times G_0}] = \frac{1}{G_0 T}\sum_{t=1}^{T}\langle \mathbf{B}^{(t)}, \mathbf{J}_{G_0\times S}\rangle.$$

The failure combination $\mathbb{S}^o = \left\{\mathcal{S}_1^o, \mathcal{S}_2^o, \ldots, \mathcal{S}_{C_S^o}^o\right\}$, where C_S^o denotes the combination symbol. The residual amount $\mathcal{A}_{\mathcal{G}_0}^{\mathcal{S}_1^o}$ of the *local access asset* $\mathcal{A}_{\mathcal{G}_0}$ after eliminating the SAT set $\mathcal{S}_1^o$ is $\mathcal{A}_{\mathcal{G}_0}^{\mathcal{S}_1^o} = \frac{1}{G_0 T}\sum_{t=1}^{T}\left\langle \mathbf{B}_{\mathcal{S}_1^o}^{(t)}, \mathbf{J}_{G_0\times S}\right\rangle$, where $\mathbf{B}_{\mathcal{S}_1^o}^{(t)} = \mathbf{B}^{(t)} \odot \mathbf{M}_{\mathcal{S}_1^o}$ denotes the ST connection matrix at moment t after masking by mask matrix $\mathbf{M}_{\mathcal{S}_1^o} \in \{0,1\}^{G_0\times S}$, $\mathbf{M}_{\mathcal{S}_1^o}$ can be given by

$$\mathbf{M}_{\mathcal{S}_1^o}(g,s) = 0 \text{ if } \mathfrak{s}_s \in \mathcal{S}_1^o \text{ else } 1, \forall g \in \{1,2,\ldots,G_0\},$$

then by analogy, $\mathcal{R}_{\mathcal{G}_0}^o$ can be derived as shown in eq. (28), shown at the top of the next page. In particular, $\mathcal{R}_{\mathcal{G}_0}^1 = (S-1)/(STG_0)\cdot\langle \mathbf{B}, \mathbf{J}_{G_0\times TS}\rangle$.

APPENDIX D
PROOF OF LEMMA 5

Vectors $\mathbf{v}_{\mathfrak{s}_i}^{(t)} = [x_{\mathfrak{s}_i}^{(t)}\ y_{\mathfrak{s}_i}^{(t)}\ z_{\mathfrak{s}_i}^{(t)}]$ and $\mathbf{v}_{\mathfrak{s}_j}^{(t)} = [x_{\mathfrak{s}_j}^{(t)}\ y_{\mathfrak{s}_j}^{(t)}\ z_{\mathfrak{s}_j}^{(t)}]$ are the WGS-84 coordinates of SAT $\mathfrak{s}_i$ and $\mathfrak{s}_j$ at moment t, which satisfies $r_{\mathfrak{s}} = ||\mathbf{v}_{\mathfrak{s}}^{(t)}||_2 = R_E + h_{\mathfrak{s}}, \forall t \in \mathcal{T}$. From Fig. 13, it can be seen that when the ISL distance is within the ISL LOS distance threshold, *i.e.* $||\mathbf{v}_{\mathfrak{s}_i}^{(t)} - \mathbf{v}_{\mathfrak{s}_j}^{(t)}||_2 \leq d_{2\mathfrak{s},\max}$, the SATs $\mathfrak{s}_i$ and $\mathfrak{s}_j$ are considered to be visible or connected at moment t. Thus according to the triangular cosine theorem $||\mathbf{d}_{\mathfrak{s}_i,\mathfrak{s}_j}^{(t)}||_2 = ||\mathbf{v}_{\mathfrak{s}_i}^{(t)}||_2 + ||\mathbf{v}_{\mathfrak{s}_j}^{(t)}||_2 - 2||\mathbf{v}_{\mathfrak{s}_i}^{(t)}|| \cdot ||\mathbf{v}_{\mathfrak{s}_j}^{(t)}|| \cdot \cos\alpha$, and ISL-available geocentric angle maximum $\alpha_{\max} = \left\{\arccos\left[1 - (||\mathbf{d}_{\mathfrak{s}_i,\mathfrak{s}_j}||_2^{\max})^2/(2||\mathbf{v}_{\mathfrak{s}}||_2^2)\right]\right\}^{-1} = \left\{\arccos\left[1 - d_{2\mathfrak{s},\max}^2/(2(R_E+h_{\mathfrak{s}})^2)\right]\right\}^{-1}$, we can easily know that the CON scale of a single-plane connected CON should satisfy $S^{\mathcal{N}}(P=1) \geq 2\pi/\alpha_{\max} \geq 2\pi/\left\{\arccos\left[1 - d_{2\mathfrak{s},\max}^2/(2(R_E+h_{\mathfrak{s}})^2)\right]\right\}^{-1}$, *i.e.* as indicating there is a CON scale lower bound $S_{\min}^{\mathcal{N}}(P=1) = \left\lceil 2\pi/\left\{\arccos\left[1 - d_{2\mathfrak{s},\max}^2/(2(R_E+h_{\mathfrak{s}})^2)\right]\right\}^{-1}\right\rceil$.

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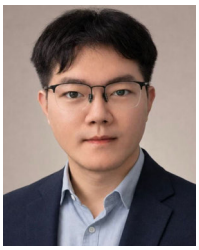
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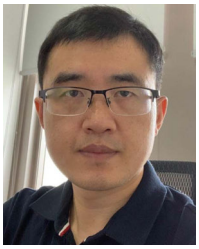
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